Provisional: descriptive bidding patterns and regulatory context

MTU15 thesis companion

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This file collects descriptive facts that any identification design must rest on: how firms bid across hour-classes (critical vs flat), how pivotal and non-pivotal firms differ, how technologies differ, and the regulatory / RT2 / reforzada context in which the reforms operate. Nothing here is an identification claim. The regression outputs from the current reform-window design are collected in a companion sibling document.

Contents

Glossary of acronyms

Acronym	Meaning
CCGT	Combined-cycle gas turbine: flexible, dispatchable thermal generator.
DA	Day-ahead market (Mercado Diario), cleared once per day.
IDA	Intraday auctions (3 European SIDC sessions post-2024-06-14).
MCP	Market clearing price.
MTU	Market time unit (MTU60 = hourly bidding; MTU15 = 15-minute bidding).
ISP15	Imbalance Settlement Period at 15 minutes (settlement granularity, started 2024-12-01).
REE	Red Eléctrica de España, the Spanish transmission-system operator.
OMIE	Operador del Mercado Ibérico de Energía, the wholesale market operator.
PDBF	Programa Diario Base de Funcionamiento — first post-DA viable schedule (REE-built).
PDBC	Programa Diario Base de Casación — pre-PDBF DA-cleared schedule (OMIE-built).
PHF	Programa Horario Final — post-IDA + post-RT viable schedule.
RES	Renewable Energy Sources (wind, solar PV, solar thermal, RoR hydro).
TR	Tiempo Real — real-time restrictions phase (PO 3.4–3.5), pay-as-bid.
Fase I/II	Restricciones técnicas Fase 1 (post-PDBF, pre-IDA) and Fase 2 (post-IDA), pay-as-bid.
aFRR	Automatic Frequency Restoration Reserve (Regulación secundaria), PICASSO platform.
mFRR	Manual Frequency Restoration Reserve (Regulación terciaria), MARI platform.
RR	Replacement Reserves (Reserva de Sustitución), TERRE platform.
BSP	Balance Service Provider (firm enrolled to supply balancing energy).
EUPHEMIA	Day-ahead pan-European auction algorithm (clears DA market).

Reform-window timeline

The five reform windows used throughout this document are defined as follows. The “pre-IDA window” (pre 2024-06-14) is the baseline regime, never directly compared in the main tables but referenced for context.

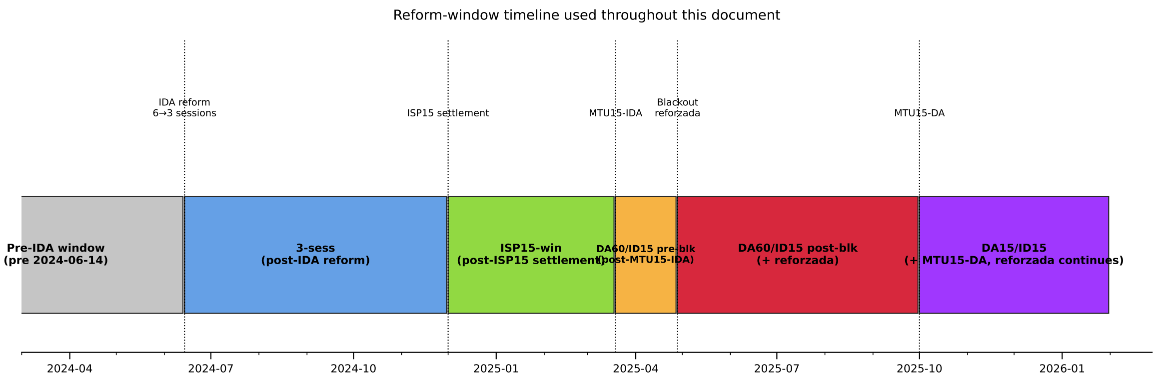


Figure 1: Reform-window timeline. Five named windows partition the post-2024-06-14 sample; the pre-IDA window is the baseline regime. Dotted vertical lines mark the five reform dates that produce the window boundaries.

Calendar coverage and same-calendar baselines

Why this matters. Spanish electricity is strongly seasonal: heating-driven demand peaks in winter, cooling in summer, hydro inflows peak in spring, solar production peaks at the summer solstice, wind blows hardest in winter. The five reform windows above do *not* span the calendar uniformly — each window covers a specific subset of calendar months, and these subsets differ across windows. Any cross-regime comparison that does not hold the calendar fixed therefore confounds the regime change with the seasonal pattern. **This applies to every variable in this document**: bid prices, D_w, Δp / Δq, redispatch volumes, imbalance MWh, system-cost levels, RES capture rates. A reader who picks any table in this document should first read Table 1 below to know which calendar months that regime covers and what comparator is available.

Table 1: Calendar coverage of each reform window, and same-calendar baseline availability. "Calendar months covered" is the set of clock months that fall inside the window. "Days" is the window length. "Same-calendar pre-IDA placebo" indicates whether a clean same-calendar baseline exists in the pre-IDA period (2018-06-14 through 2024-06-13), where no granularity reform was active. "Reforzada overlay" flags windows in which the post-blackout reinforced-operation policy (active 2025-04-28 onward) is also in effect.

Regime	Calendar dates	Months covered	Days	Reforzada?	Same-calendar pre-IDA placebo
pre-IDA	$\leq$ 2024-06-13	all 12, multi-year	–	no	N/A (is the baseline)
3-sess	2024-06-14 – 2024-11-30	Jun–Nov	170	no	Clean (Jun–Nov 2018–2023)
ISP15-win	2024-12-01 – 2025-03-18	Dec–Mar	108	no	Clean (Dec–Mar 2018–2024)
MTU15-IDA pre-blk	2025-03-19 – 2025-04-27	late Mar – Apr	40	no	Clean (same 40 days 2018–2024)
MTU15-IDA post-blk	2025-04-28 – 2025-09-30	May – Sep	156	**yes**	Clean for granularity (May–Sep 2018–2023, pre-IDA); none for reforzada (no pre-2025-04-28 reforzada in data)
DA15/ID15	2025-10-01 – present	Oct–May (partial, ongoing)	220+	**yes**	None clean. 2024–25 same-calendar comparators are themselves a mix of post-MTU15-IDA + ISP15-win regimes

Three categories. Each regime falls into one of three buckets for the purpose of cross-regime reading:

1. **Clean same-calendar placebo available.** *3-sess, ISP15-win, MTU15-IDA pre-blk.* For each of these, we can restrict the pre-IDA baseline to the same calendar months in 2018–2024 (and average across years), and the difference (regime − pre-IDA same-cal) is interpretable as the regime effect with seasonality differenced out. Any cross-regime reading that uses these windows defaults to this same-calendar comparison unless explicitly stated otherwise.

2. **Reforzada-confounded, partial calendar-match available.** *MTU15-IDA post-blk.* The window spans May–Sep 2025, which has clean same-calendar comparators in pre-IDA (May–Sep 2018–2023). But these comparators are pre-reforzada. So the difference (MTU15-IDA post-blk − pre-IDA same-cal) confounds (a) the granularity reform, (b) operación reforzada, and (c) the post-blackout system state. There is no calendar period in the data in which reforzada is active but MTU15 is not, so reforzada cannot be differenced out via a same-calendar pre-reform window. For descriptive purposes, May–Sep 2025 is best read as "post-MTU15-IDA + reforzada + post-blackout, jointly"; see §6 on disentangling these channels.

3. **No clean calendar match.** *DA15/ID15.* The window covers Oct 2025 onward. The same calendar months one year earlier (Oct 2024 – May 2025) are themselves a mix of pre-MTU15-IDA, ISP15-win, MTU15-IDA pre-blk, and MTU15-IDA post-blk regimes — so calendar-matching to "one year ago" does not produce a single-regime comparator. The cleanest available calendar comparators for DA15/ID15 are the same months in pre-IDA years (Oct 2023 – May 2024 was largely pre-IDA, with the IDA-3 reform starting only mid-Jun 2024; Oct 2022 – May 2023 and earlier are all pre-IDA). But even with pre-IDA same-calendar baselines, the reforzada overlay is unresolved by this comparator: DA15/ID15 is the union of MTU15-DA, MTU15-IDA, and reforzada in the same calendar window, and our data does not contain a period where any one of these three is active without the others. Therefore any DA15/ID15 reading should be flagged as descriptive only, with the three confounds explicitly named.

Fourier-deseasonalization. For a daily outcome y_t we fit a pooled Frisch-Waugh-Lovell regression on $t \in [2022\text{-}01\text{-}01, 2026\text{-}05\text{-}15]$:

$$g(y_t) \;=\; \alpha \;+\; \sum_r \beta_r \,\mathbf{1}\{\text{regime}(t) = r\} \;+\; \sum_{k=1}^{4} \left[a_k\, c_k(t) + b_k\, s_k(t)\right] \;+\; \sum_{j=1}^{6} \delta_j \,\mathbf{1}\{\text{dow}(t) = j\} \;+\; \varepsilon_t,$$

with omitted regime pre-IDA, link g equal to log for positive levels and logit for bounded shares, and day-of-year sinusoids $c_k(t) = \cos(2\pi k \ \text{doy}(t)/365)$, $s_k(t) = \sin(2\pi k \ \text{doy}(t)/365)$. Four harmonics (annual, semi-annual, four-month, quarterly) capture the seasonal cycle; the six day-of-week dummies absorb weekly periodicity. Seasonality is pooled across regimes because the 40-day MTU15-IDA pre-blk window cannot identify regime-specific Fourier coefficients.

Seasonality-adjusted (SA) value. $\hat{y}_r^{\text{SA}}$ is the predicted level of y in regime r at annual-mean Fourier (zero by orthogonality) and within-week DOW mean, with a Duan-smearing correction across in-sample residuals $\{\widehat{e}_i\}$ for the non-linear link:

$$\hat{y}_r^{\text{SA}} \;=\; \frac{1}{n}\sum_{i=1}^{n} g^{-1}\left(\widehat{\alpha} + \widehat{\beta}_r + \tfrac{1}{7}\sum_{j=1}^{6} \widehat{\delta}_j + \widehat{e}_i\right).$$

In tables, SA values sit in coloured brackets next to the raw regime mean (omitted where the value is calendar-matched by construction). Where raw and SA move the same way the pattern survives seasonality control; where they diverge, only the SA value is comparable across regimes. For the reforzada-confounded windows (MTU15-IDA post-blk, DA15/ID15) the SA value still bundles granularity, reforzada, and the post-blackout system state — it differences out the calendar, not the policy overlap.

Reading guide. Part A is about bidding and trading patterns. Part B is about the regulatory/post-clearing channels (reforzada, geographic dominance, CNMC sanctions, etc.) that are independent of the granularity reform but constrain how the identification can be set up.

Part I
Part A — Bidding and trading patterns

Reading guide for Part A. Three setup sections come first. §1 defines the **hour-class taxonomy** (critical / flat / midday) used to slice every table and figure that follows. §2 characterises **who price-sets** in the day-ahead market (§2.1) and, methodologically, in the intraday markets (§2.2). §3 then defines the **bid-shape metrics** in two families: *per-curve functionals* (one scalar per single bid curve — σ_p, N_{eff}, the fPCA scores) and *pairwise divergences* (a statistic of two curves — D_w, Δp, Δq). Visual results (per tech × firm × quarter bid curves with mean clearing prices overlaid) come in §3.3; quantitative tables come in §3.4.

1 Hour-class taxonomy: critical, flat, midday

Classifier: within-hour standard deviation of residual demand. MTU15 granularity has economic content only where the within-hour profile actually moves: if residual demand is flat within a clock-hour the hourly clearing price already summarises it, and granular pricing adds no signal; if it ramps sharply (morning ramp, evening peak, solar fade) granular pricing unlocks quarter-specific discrimination. We therefore classify hours by within-hour *variability*, not demand level. For clock-hour h, date d, residual demand $D_{h,q,d}$ = load − wind − solar at quarter q with quarter-mean $\bar{D}_{h,d}$,

$$\sigma_{\text{within}}(h) \;=\; \mathbb{E}_d\left[\sqrt{\tfrac{1}{4}\sum_{q=1}^{4}\big(D_{h,q,d}-\bar{D}_{h,d}\big)^2}\,\right] \quad \text{[MW]},$$

computed from ENTSO-E A65 load and A75 wind+solar at 15-min ISP over 2023-01-01 to 2026-05-14 (the longest clean 15-min window). High σ_{within} marks **critical** hours, where strategic within-hour bidding has scope; low marks **flat** hours, where it has none — so flat is the within-day control, and the (critical − flat) differential strips any day-level shock that hits both classes proportionally.

Table 2: Hour-of-day classification metrics, Spain 2023–2026 (ENTSO-E A65 + A75, 15-min ISP, $n \approx 3.4$ years of daily observations per hour). The classifier is σ_{within} (column 6, MW). Critical hours: $\sigma_{\text{within}} \in [522, 1303]$. Flat hours: $\sigma_{\text{within}} \in [336, 357]$. Midday: $\sigma_{\text{within}} \in [468, 502]$ — intermediate, but excluded for the separate reason discussed below. Dropped (transition) hours sit at the regime boundaries. *Year-stability:* across the 3 full years 2023, 2024, 2025 (computed separately), the critical-set min σ_{within} across years is 530 MW, exceeding the flat-set max (387 MW) by 1.4×; no within-class swaps across years.

Clock-hour	Demand (GW)	Solar (GW)	Wind (GW)	Residual demand (GW)	σ_{within} (MW)	CV (%)	Δ demand (MW)	Class
00:00–01:00	22.8	0.1	7.3	15.4	412	2.7	−834	*dropped*
01:00–02:00	22.0	0.1	7.1	14.8	347	2.4	−470	*flat*
02:00–03:00	21.6	0.1	6.9	14.6	336	2.3	−190	*flat*
03:00–04:00	21.5	0.1	6.7	14.7	357	2.4	176	*flat*
04:00–05:00	22.4	0.1	6.7	15.7	587	3.8	1114	*dropped*
05:00–06:00	24.4	0.2	6.7	17.5	721	4.1	1743	*critical*
06:00–07:00	26.5	1.5	6.6	18.4	915	5.0	1423	*critical*
07:00–08:00	27.7	4.9	6.2	16.6	1078	6.5	627	*critical*
08:00–09:00	28.3	9.1	5.8	13.4	1006	7.5	253	*critical*
09:00–10:00	28.3	12.0	5.5	10.9	753	6.9	−104	*dropped*
10:00–11:00	28.2	13.3	5.4	9.5	499	5.3	−66	*dropped*
11:00–12:00	28.2	13.8	5.5	8.9	468	5.3	94	*dropped*
12:00–13:00	28.3	13.8	5.7	8.8	502	5.7	−104	*dropped*
13:00–14:00	27.9	13.5	5.8	8.5	485	5.7	−331	*dropped*
14:00–15:00	27.5	12.9	6.1	8.5	469	5.5	−122	*dropped*
15:00–16:00	27.6	11.7	6.4	9.6	677	7.1	291	*dropped*
16:00–17:00	28.2	9.0	6.8	12.3	1109	9.0	553	*critical*
17:00–18:00	29.2	5.4	7.4	16.4	1303	7.9	891	*critical*
18:00–19:00	30.5	2.2	7.7	20.5	1065	5.2	884	*critical*
19:00–20:00	31.3	0.6	7.9	22.8	522	2.3	176	*critical*
20:00–21:00	30.4	0.3	7.9	22.2	622	2.8	−1444	*critical*
21:00–22:00	28.0	0.2	7.9	20.0	781	3.9	−1851	*critical*
22:00–23:00	25.8	0.2	7.7	18.0	664	3.7	−1407	*critical*
23:00–00:00	24.2	0.1	7.5	16.5	510	3.1	−1062	*dropped*

Classification rule. Read off Table 2 by contiguous-block structure: **critical** = morning ramp (5h–8h) + evening peak (16h–22h), 11 hours, $\sigma_{\text{within}} \in [522, 1303]$ MW; **flat** = overnight (1h–3h), 3 hours, $[336, 357]$ MW; **midday** = solar peak (11h–14h), 4 hours, intermediate σ_{within} but residual demand < 9 GW. Midday is excluded from the control set: its low residual demand shifts the marginal technology from CCGT/hydro to solar PV + storage, so pooling it into flat would absorb a separate operational channel. The 6 boundary hours (0, 4, 9, 10, 15, 23) straddle the thresholds and are dropped — a 14-hour analytic sample (11 critical + 3 flat). The CV column ($\sigma_{\text{within}}/\overline{\text{residual demand}}$) flags that "critical" is not uniform: evening-peak hours carry a large MW swing on a high demand base (low CV $\sim$ 2–4%), morning-ramp hours a large swing on a thin base (high CV $\sim$ 8–9%, more relative room for a within-hour price-setter).

Validation. Two checks. (i) *No cost-shock confound*: CCGT price-setting share is 9.1% critical vs 9.0% flat — gas-cost shocks cannot propagate asymmetrically across the two classes, so within-day differencing identifies behaviour, not cost. (ii) *Outcome-side*: post-MTU15-DA the within-hour DA quarter clearing-price SD averages 6.58 EUR/MWh in critical hours vs 2.13 in flat (3.1×), so the classifier correctly predicts where post-reform quarter prices actually diverge.

2 Who price-sets

2.1 For the day-ahead market

Method. OMIE submits stepwise orders to EUPHEMIA: in-the-money fully accepted, out-of-the-money rejected, **at-the-money curtailable**. The price-setter is the unit whose at-MCP step is *partially* accepted: with q_{below} the MW bid strictly below MCP, q_{at} the MW at MCP (± 0.01 EUR/MWh), and q_{assigned} the unit's cleared MWh,

$$q_{\text{below}} < q_{\text{assigned}} < q_{\text{below}} + q_{\text{at}}.$$

Full acceptance (inframarginal) and curtailed-to-zero (often a minimum-income deactivation) are excluded. A naive rule flagging every $p_{\text{bid}} = \text{MCP}$ — the CNMC market-supervision metric — pools all three cases and is mechanically larger: for CCGT DA15/ID15 critical it gives 9.1% against our 2.7% (most CCGT at-MCP steps are minimum-income-deactivated to zero). Aggregations are MW-weighted by q_{at}.

Table 3: Price-setter shares by technology and hour-class, across 5 reform-window regimes (sell-side partial-acceptance, MW-weighted by q_{at}). C = Critical, F = Flat. Caveat: DA60/ID15 pre-blackout is 40 days only — directional.

	3-sess		**ISP15-win**		**DA60/ID15 pre**		**DA60/ID15 post**		**DA15/ID15**	
Tech	C	F	C	F	C	F	C	F	C	F
Wind	28.3	17.9	37.8	63.9	47.5	70.9	29.6	17.3	29.3	53.0
Solar PV	17.7	0.0	1.4	0.0	20.6	0.0	29.9	0.0	7.7	0.0
Hydro pump	10.4	13.3	17.0	2.7	2.2	10.8	8.0	19.3	28.4	13.0
Hydro	11.9	36.1	19.6	10.9	2.7	1.7	9.4	21.5	23.7	20.0
Nuclear	12.5	20.2	4.2	6.5	15.6	13.5	7.9	5.5	4.4	3.3
CCGT	8.0	8.9	17.2	11.1	0.8	0.0	6.2	33.1	2.7	2.0
Cogen	3.0	0.3	0.1	0.1	2.5	0.1	2.1	0.1	0.5	0.9
Other RES	3.9	1.5	0.6	0.2	3.4	0.8	3.1	1.2	1.2	2.2
Import	0.8	0.8	0.5	3.1	0.0	0.0	0.0	0.0	0.5	3.3

Sell-side price-setter shares by technology and regime.

CCGT firm-level: GN never price-sets, IB and GE swap dominance. Within the CCGT column above, who owns the at-MCP step?

Regime	Hour-class	IB	GE	GN	HC	OTHER
3-sess	Critical	**60**	31	0	0	8
3-sess	Flat	16	**50**	0	0	25
ISP15-win	Critical	35	**57**	0	0	5
ISP15-win	Flat	2	**74**	0	3	20
DA60/ID15 post-blk	Critical	31	**69**	0	0	0
DA60/ID15 post-blk	Flat	19	**80**	0	0	0
DA15/ID15	Critical	**65**	29	0	6	0
DA15/ID15	Flat	**80**	0	6	14	0

Table 4: Per-firm shares within the CCGT price-setter pool (% of CCGT q_{at}). **GN's CCGT bid stack is so heavily concentrated at scarcity prices ($\geq$ 500 EUR/MWh) that GN essentially never lands at MCP**. IB and GE alternate as the dominant CCGT price-setter. The 33.1% CCGT flat-hour figure in DA60/ID15 post-blackout is 80% GE.

Reading across regimes.

- **Wind is the largest sell-side price-setter** almost everywhere (18–71% MW share): RE aggregator portfolios systematically land a single block at-the-money. **Iberdrola**

dominates the marginal MW in DA15/ID15 (top-firm share ≈ 65% critical, 80% flat) through pump-storage (MUEL), hydro and the IBEVD11 wind aggregator.

- **CCGT's DA price-setter share is small and fluctuates**: 8% → 17% → 0.8% → 6% (critical). The lone spike — 33% flat under DA60/ID15 post-blk — is 80% Endesa, two units submitting single full-nameplate tranches at MCP.
- **GN does not partial-accept at MCP but is strategically active within-hour**: its CCGT bid mass sits at scarcity prices (≥ 500 EUR/MWh), yet its within-quarter bid *shape* is the largest of any CCGT firm (§3.4).
- **Cross-regime CCGT symmetry** (a gas-cost-shock falsifier for flat vs critical) holds in 3-sess (8.0 vs 8.9) and DA15/ID15 (2.7 vs 2.0) but breaks in ISP15-win and DA60/ID15 post-blk.

2.2 For the intraday markets

Method. Identical partial-acceptance rule, applied to the intraday-auction clearing prices, bid stack and programs (sell side, pooled across the 3 IDA sessions per day).

Table 5: IDA price-setter shares by technology, hour-class, and regime (at-the-money partial-acceptance, sell side), MW-weighted by q_{at}. C = Critical, F = Flat. Shares sum to 100 per (regime, hour-class).

	3-sess		**ISP15-win**		**DA60/ID15 pre**		**DA60/ID15 post**		**DA15/ID15**	
Tech	C	F	C	F	C	F	C	F	C	F
Wind	35.5	15.7	24.8	40.2	52.6	53.8	37.8	18.8	31.9	42.4
CCGT	14.1	29.2	16.1	15.3	3.1	2.8	9.7	23.1	**24.2**	**32.6**
Hydro pump	12.9	23.4	17.2	2.2	2.5	13.0	6.2	26.1	14.0	3.8
Solar PV	10.5	0.0	1.7	0.1	20.3	0.0	22.8	0.0	7.5	0.1
Hydro	6.4	15.9	18.4	23.4	4.4	3.4	5.6	15.6	10.0	10.6
Retailer	10.2	6.2	10.4	12.1	5.3	7.5	7.3	11.8	5.7	5.2
Nuclear	2.7	3.2	0.0	0.0	1.0	13.8	2.5	1.3	2.5	0.0
Other RES	1.0	1.1	1.7	0.0	3.1	0.7	2.4	0.2	0.6	0.4

Reading. **CCGT is far more active as an IDA price-setter than as a DA one**: 24.2% of critical-hour IDA mass under DA15/ID15 against 2.7% in DA — a factor of ∼ 9. The natural reading is DA-to-IDA strategic substitution: CCGTs bid above the DA clearing price (the q_1-drop story, §5) and re-enter through the intraday auction, where they systematically clear at the marginal step. Wind still dominates (∼ 32% critical) but by a smaller margin than in DA.

3 Bid-curve shape: per-curve functionals and pairwise divergences

We measure bid-curve shape with two conceptually distinct families of statistics, and keep them separate throughout.

- **Family 1 — per-curve functionals.** A functional maps a *single* bid curve to a scalar, $T : f \mapsto T(f) \in \mathbb{R}$; no comparison between curves enters its definition. Cross-regime, cross-firm and cross-hour reading is then done by comparing the *values* $T(f)$ across curves. This family holds the price spread σ_p and the effective tranche count N_{eff} (§3.1), and the functional-PCA level/tilt scores ξ_{ik} of §3.5 — the score $\xi_{ik} = \langle f_i - \bar{f},\, \phi_k \rangle$ is itself a per-curve functional, a shape coordinate of the single curve f_i. **This is the primary descriptive object**: it asks what one bid curve looks like, and how that value moves across regimes.

- **Family 2 — pairwise divergences.** A divergence is a statistic of *two* curves, $D(f_i, f_j)$: it does not summarise either curve on its own, it measures how far apart they are. This family holds D_w and its Δp / Δq channel decomposition (§3.2). We use it for one comparison only — a unit's four within-hour quarter curves against each other — as a within-hour-shaping diagnostic, secondary to Family 1.

The distinction is the one the advisor asked for: characterise each bid curve by its own value (Family 1), rather than by divergences between curves (Family 2). Family 2 is retained only because the within-hour quarter comparison has no single-curve analogue.

3.1 Family 1 — per-curve functionals: σ_p and N_{eff}

The strategic band, chosen data-driven. Both families restrict to a price band $|p - \text{MCP}| \leq h$ — the region where a bid is in contention — and we read h off the data rather than assert it. Figure 2 plots the MW-weighted density of the bid-to-clearing distance $|p_{\text{bid}} - \text{MCP}|$ over all day-ahead sell tranches of the price-setting techs. It is **bimodal**: a dense near-MCP *competing* cluster and a sparse far *scarcity-cap* cluster (withholding bids near the price cap). We set h at the **antimode** — the first local minimum past the competing cluster. That cluster ends in a sharp cliff near 130 EUR/MWh and the valley bottoms at $h = 230$; this band holds 62% of sell-bid MW and excludes the cap cluster by construction, where the legacy $h = 50$ kept only $\sim$ 26% of CCGT sell-bid MW and understated every shape statistic. h is estimated once — pooled across regimes and techs — and frozen; a regime- or tech-specific band would move with behaviour and contaminate the cross-comparisons.

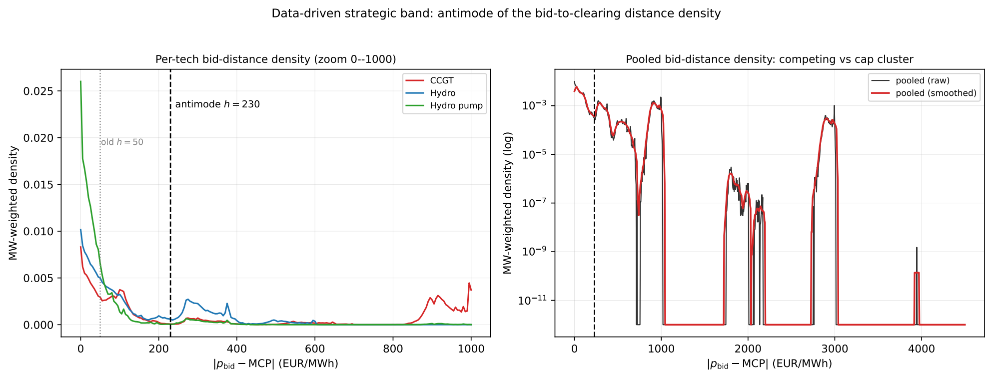


Figure 2: Data-driven strategic band. MW-weighted density of $|p_{\text{bid}} - \text{MCP}|$ for day-ahead sell tranches (price-setting techs, all regimes pooled). Left: per-tech, zoomed to 0–1000 EUR/MWh. Right: pooled, log scale — the near-MCP competing cluster and the scarcity-cap cluster show as two modes. The band edge $h = 230$ (dashed) sits at the antimode between them; the legacy $h = 50$ (dotted) fell deep inside the competing cluster.

The two-tier bid curve. The bimodality of Figure 2 is, unit by unit, a **two-tier bid curve**. Figure 3 aggregates each Big-4 CCGT fleet's day-ahead sell tranches into one supply curve: a low *competing tier*, priced near marginal cost and offered near-inelastically (it clears the auction whatever the clearing price), and a high *scarcity tier* parked far above any plausible clearing price — which does not clear, but is the capacity REE recalls in pre-IDA Fase I redispatch, paid pay-as-bid (§6). The split is sharp and firm-specific: Naturgy is the textbook case — $\sim$ 9% of its offered MW clears the auction and 75% sits in a flat shelf at $\sim$1,000 EUR/MWh — Endesa parks $\sim$22% (much of it at the 4,000 price cap) and EDP-HC $\sim$19%, while Iberdrola bids a single competing tier near marginal cost with no scarcity shelf. This is why the band is the right object: the per-curve functionals are computed on the competing tier (inside the band), where price-graded bidding is a genuine strategic choice; the scarcity tier is a separate withholding margin, excluded by construction.

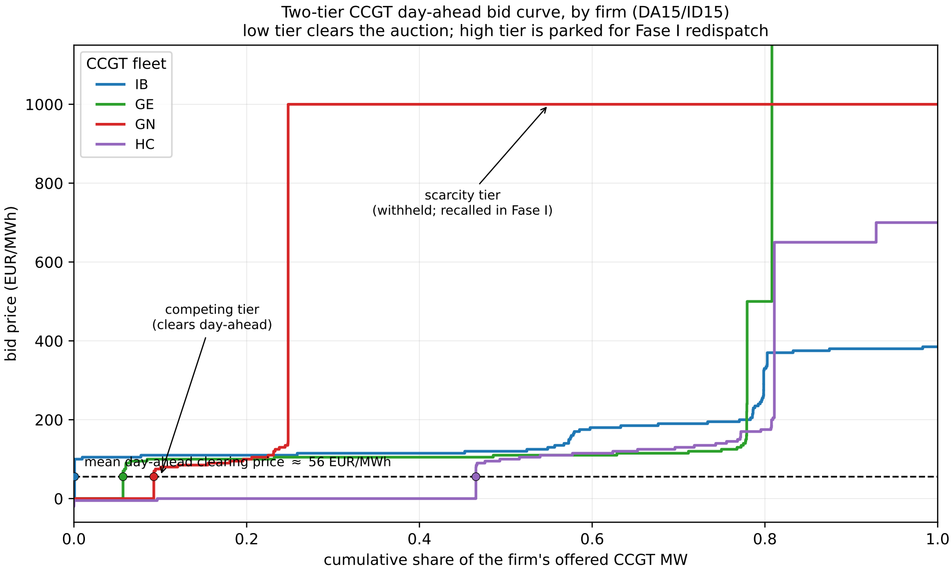


Figure 3: Two-tier CCGT day-ahead bid curve, by firm (DA15/ID15). Each Big-4 CCGT fleet's day-ahead sell tranches aggregated into one supply curve: bid price against cumulative share of offered MW. The low competing tier sits near the mean clearing price (dashed) and clears the auction; the high scarcity tier does not clear and is the capacity recalled in Fase I. Naturgy parks 75% of its MW in a flat shelf at ~1,000 EUR/MWh; Endesa's scarcity tier runs off the top of the axis (it includes ~19% of its MW at the 4,000 price cap); Iberdrola bids no scarcity tier.

Price spread σ_p and effective tranche count N_{eff}. For each (unit, date, market-period) sell bid curve, restricted to the band $|p - \text{MCP}| \leq h$, we compute two scalars:

- **Price functional — price spread σ_p**: the MW-weighted standard deviation of the in-band tranche prices (EUR/MWh). It measures how much the unit *varies its price* across the MW it offers near the clearing price. A single flat block $\Rightarrow \sigma_p = 0$; a graded ladder spanning the band $\Rightarrow \sigma_p$ large.
- **Quantity functional — effective tranche count N_{eff}**: the inverse Herfindahl of the tranche MW shares, $N_{\text{eff}} = (\sum_t q_t)^2 / \sum_t q_t^2$. It measures how the unit *splits its quantity*: $N_{\text{eff}} = 1$ means all in-band MW sits in one block, larger N_{eff} means the MW is divided into more comparable pieces.

Both are per-curve and require no curve-to-curve comparison, and are computed on the *raw* bid curve (seasonality is handled at the aggregation step, by reading the values within calendar-matched windows). The two decouple: a unit can split into many tranches at nearly one price (high N_{eff}, low σ_p) or place two blocks far apart (low N_{eff}, high σ_p). The fPCA scores of §3.5 are the third member of this family — a multivariate per-curve functional that lets the data choose the modes of shape variation rather than fixing them a priori.

Offer types: what the functionals see. The functionals are computed on the price-quantity tranche ladder a unit submits for the period — the *simple offer*, in the day-ahead offer-detail record. On top of it a unit may declare *complex conditions* on the offer header: a minimum-income condition (the unit is dispatched only if its revenue clears a fixed-plus-variable floor), indivisibility of the first tranche, a load gradient (an inter-period ramp limit), or a scheduled stop; the intraday auctions additionally admit *block orders* (one price spanning several periods, accepted whole). This is not innocuous for the metric: σ_p and N_{eff} read the submitted geometry, and the *same value* can mean different things across offer types. A near-zero σ_p is genuine flat bidding for a pure simple offer, but for a minimum-income unit it can instead reflect strategy displaced onto the income condition — a flat ladder, with dispatch governed by the income floor rather than the price steps. An indivisible first tranche is a quantity lump, not a graded step, so it depresses N_{eff} for a structural rather than behavioural reason. A block order contributes a single price to each period it covers, mechanically pushing $\sigma_p \to 0$ and $N_{\text{eff}} \to 1$. The functionals are cleanest for simple offers; for complex or block offers they still describe the curve faithfully, but the step from geometry to *strategic shaping* must be read conditional on the offer type.

3.2 Family 2 — pairwise divergences: D_w, Δp, Δq

D_w, in full. For a cell (firm, unit, date, clock-hour) the hour splits into 4 quarters. In quarter q the unit's band-restricted sell curve is the right-continuous step function $S_q(p) = \sum_k x_{qk}\,\mathbf{1}\{p_{qk} \le p\}$ on $p \in [c-h, c+h]$, with $S_q(c-h) = 0$ and $S_q(c+h) = M_q$ (the band-mass), where c is the hour-mean of the four quarter clearing prices — one reference common to all 6 quarter-pairs, so the band does not move across quarters. For two quarters i, j the kernel-weighted Wasserstein-1 dissimilarity is

$$D_w^{(i,j)} \;=\; \int_{c-h}^{c+h} K_h(p-c)\,\big|S_i(p) - S_j(p)\big|\,dp, \qquad K_h(u) = \max\big\{0,\ 1-(u/h)^2\big\}\,,$$

with K_h the Epanechnikov kernel down-weighting bids far from clearing; D_w has units EUR. The cell statistic is the mean over the 6 quarter-pairs (for the binary flag $D_w > 0$, mean, max and "any pair" agree).

Price and quantity channels. Splitting the underlying Wasserstein-1 distance at the common mass $\min(M_i, M_j)$ separates a price channel from a quantity channel,

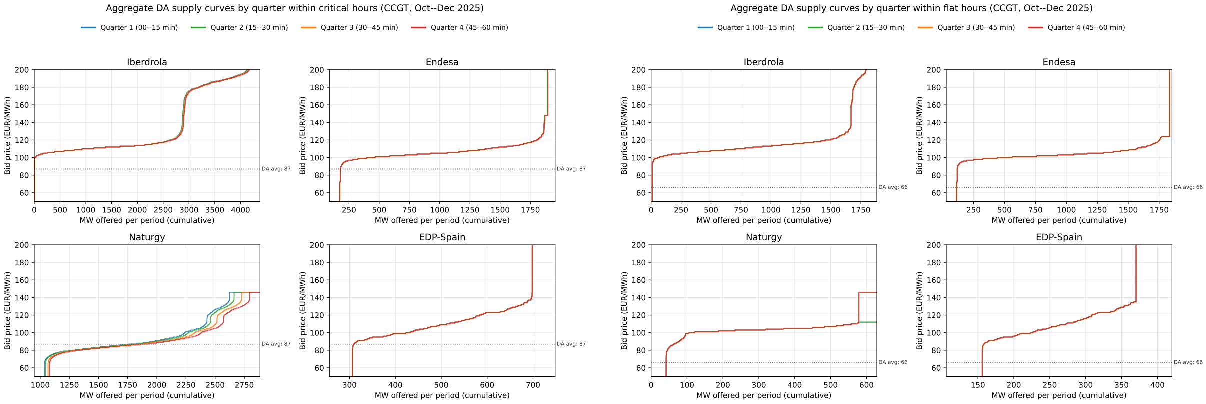
$$\Delta q_{ij} = |M_i - M_j| \;\;\text{[MWh]}, \qquad \Delta p_{ij} = \frac{1}{\min(M_i, M_j)} \int_0^{\min(M_i,M_j)} \big|S_i^{-1}(Q) - S_j^{-1}(Q)\big|\,dQ \;\;\text{[EUR/MWh]}.$$

Δp and Δq are read as channel *indicators* (active/inactive, and which dominates), not additive contributions — raw Δq in MWh is scale-dependent, so we lean on the active/inactive distinction.

Band, and a pending refresh. D_w, Δp and Δq use the same data-driven band as Family 1 ($h = 230$, §3.1). *Pending:* the D_w / Δp / Δq tables in §3.4 are still at the legacy $h = 50$ — a $\sim 4.6\times$ wider band can saturate the binary $D_w > 0$ flag, an analytical change held as a separate step rather than a mechanical re-run.

3.3 Bid graphs by tech, firm, and within-hour quarter

The figures below show the aggregate within-hour quarter bid curves with the mean clearing price overlaid (so the strategic band $|p-c| \le 230$ is visible). Aggregation across ~ 92 DA15/ID15 dates per (firm, hour, quarter) cell smooths per-day shaping — the per-curve functionals of §3.4 are the systematic measure, these are the visual companion. *Reading*: same-quarter overlap in flat hours is the falsification; visible quarter spread in critical hours is the strategic signal, concentrated in GN CCGT and Endesa Hydro. The IDA-side and pump-storage-demand analogues are similar and not reproduced.



(a) Critical hours (05:00–09:00, 16:00–23:00) **(b)** Flat hours (01:00–04:00) — falsification

Figure 4: Aggregate DA supply curves by within-hour quarter, CCGT per firm, Oct–Dec 2025. Quarter overlap in flat hours is essentially complete (the falsification); critical-hour quarter spread is visible for GN.

3.4 Results: per-curve functionals, and the pairwise D_w diagnostic

The results below quantify what the bid-curve figures in §3.3 show visually: first the Family-1 per-curve functionals (σ_p, N_{eff}, defined in §3.1), then the Family-2 pairwise D_w diagnostic (§3.2).

Per-curve functionals: price spread and effective tranche count. Table 6 and Figure 5 report σ_p and N_{eff} averaged over (unit, date, period) day-ahead sell-bid curves, by tech, hour-class and regime — the cross-regime reading is a comparison of per-curve *values*, with no differencing of curves.

Table 6: Per-curve bid-shape metrics by tech, hour-class and regime (DA sell bids, in-band $|p - \text{MCP}| \leq 230$ — the data-driven band of §3.1 — mean over (unit, date, period) curves). σ_p = MW-weighted price spread (EUR/MWh); N_{eff} = effective in-band tranche count. 994,616 in-band curves.

		3-sess		ISP15-win		DA60/ID15 pre		DA60/ID15 post		DA15/ID15	
Tech	Hour	σ_p	N_{eff}	σ_p	N_{eff}	σ_p	N_{eff}	σ_p	N_{eff}	σ_p	N_{eff}
CCGT	Critical	40.5	3.2	51.4	3.3	35.1	3.3	28.0	3.1	28.6	3.4
CCGT	Flat	27.3	2.6	44.8	2.7	21.3	2.4	20.2	2.6	19.1	2.3
Hydro	Critical	18.2	3.6	18.8	3.5	16.1	3.6	18.4	3.5	21.6	3.8
Hydro	Flat	18.0	3.6	18.3	3.4	16.4	3.5	19.0	3.5	20.1	4.0
Hydro pump	Critical	3.8	2.4	4.0	2.4	3.5	2.3	4.3	2.8	4.7	3.3
Hydro pump	Flat	2.1	2.5	3.8	2.8	3.7	2.3	2.8	2.5	2.9	3.8

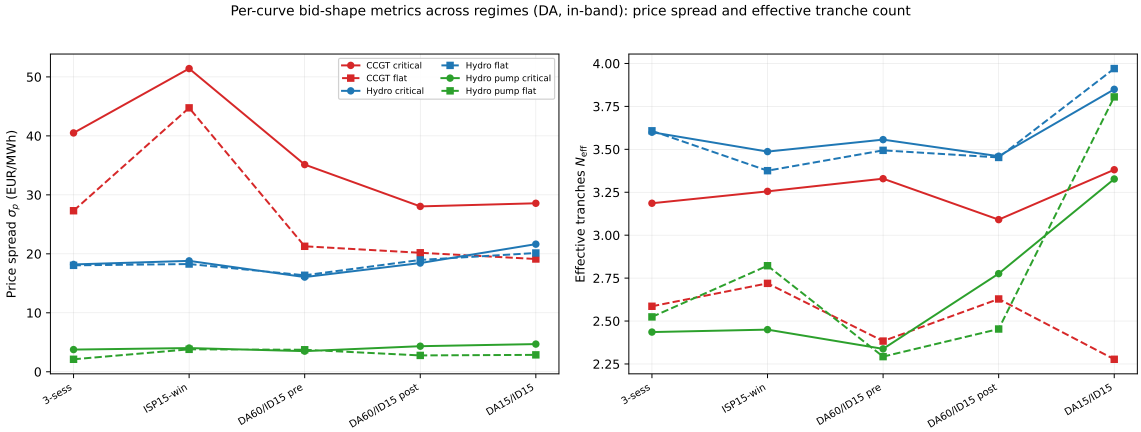


Figure 5: Per-curve bid-shape metrics across regimes (DA, in-band, data-driven band $|p - \text{MCP}| \leq 230$). Left: price spread σ_p; right: effective tranche count N_{eff}. Each line is a (tech, hour-class) cell; solid = critical hours, dashed = flat. CCGT carries the widest price spread ($\sigma_p \approx$ 28–51 EUR/MWh) and it is consistently larger in critical than flat hours in every regime — the strategic signal; Hydro is intermediate ($\sim$ 16–22) and Hydro pump near-flat ($\sim$ 4). N_{eff} sits at $\sim$ 3–4 for the price-setting techs, rising mildly for Hydro pump across the reform sequence.

The pairwise D_w diagnostic (Family 2). D_w asks whether a unit's four within-hour quarter curves *differ* near clearing; the per-cell binary flag $D_w > 0$ is the within-hour-shaping signal. The cleanest reform-window test is the IDA blackout split for CCGT — pre-blackout (2025-03-19→04-27, MTU15-IDA active, reforzada-free) vs post-blackout (04-28→09-30, reforzada). If REE post-clearing intervention removed the leverage on the clearing price, pre > post.

Table 7: IDA quarter-dissimilarity, pre vs post-blackout, CCGT. Percentage of (firm, unit, date, hour) cells with $D_w > 0$ (i.e., quarters differ within the Epanechnikov band, $h = 50$ EUR/MWh, around hour-mean IDA clearing price). *Pre-blackout*: 2025-03-19 → 2025-04-27 (40 days, MTU15-IDA active, reforzada-free). *Post-blackout*: 2025-04-28 → 2025-09-30 (155 days, reforzada active).

	Pre-blackout			Post-blackout		
Firm	Critical	Flat	Crit−Flat	Critical	Flat	Crit−Flat
IB	9.6	7.0	+2.6	20.3	6.2	+14.1
GE	14.1	10.0	+4.1	43.8	33.2	+10.6
GN	30.5	9.2	+**21.3**	53.8	37.9	+15.9
HC	27.9	20.7	+7.2	34.2	21.2	+13.0

Reading. The directional prediction (pre > post) is *rejected* — $D_w > 0$ rates roughly double post-blackout for every firm. But the critical-vs-flat asymmetry survives and broadens: pre-blackout only GN shows a clean gap (+21pp), post-blackout all four firms do (+11 to +16pp). The strategic within-hour mechanism is essentially unchanged from pre-reforzada; the relevant new lever is geographic (§7). A complementary $\Delta p/\Delta q$ split (DA, CCGT) shows GN shaping is quantity-only (18% of cells Δq-active, $\sim$ 0% Δp-active) — the Cournot version of price-raising: a fixed near-cap price ladder with quarter-to-quarter quantity reallocation.

Within-hour shaping by firm and unit. A per-unit $\Delta p/\Delta q$-activity scan of DA15/ID15 critical hours confirms within-hour shaping is exploited by specific firm-unit combinations, not uniformly. GN never price-sets at MCP (its bids are too high, §2.1) yet shapes within-hour bids in 33% of critical hours with the largest median Δp of any CCGT firm; Iberdrola is the most active shaper across CCGT (68% of cells Δp-active), hydro (74%) and pump-storage (88%);

among wind aggregators Gesternova ladder-bids across quarters (96% Δp-active) while AXPO and NEXUS bid identically across all four.

3.5 Functional PCA of bid-curve shape changes across regimes

Method. The two functionals of §3.1 fix the shape coordinate in advance; functional PCA (fPCA) is the data-driven member of Family 1 — it maps each single curve to scalars but lets the data choose the modes. Each per-(unit, date, period) bid stack is the inverse supply curve $f_i(q) = S_i^{-1}(q)$ on cumulative-MW quantiles $q \in [0,1]$; with sample mean $\bar{f}$, fPCA finds an orthonormal basis $\{\phi_k\}$ with uncorrelated scores

$$\xi_{ik} \;=\; \langle f_i - \bar{f},\, \phi_k \rangle_{L^2} \;=\; \int_0^1 \big[f_i(q) - \bar{f}(q)\big]\, \phi_k(q)\, dq,$$

so the Karhunen-Loève expansion $f_i = \bar{f} + \sum_k \xi_{ik}\phi_k$ replaces each curve by K scalar scores — PC1 a level shift, PC2 a cheap-vs-scarcity tilt, PC3 finer shape. We pool per tech, deseasonalise functionally before PCA, and isolate each reform with a pairwise pre/post score regression on the SA scores (per tech, firm, PC). Table 8 reports each pre-post PC change projected back onto the curve via $\widehat{\beta}_k\phi_k$ (EUR/MWh). PC1 carries most of the per-tech variation (explained-variance 0.86–0.91 for CCGT/Hydro/Hydro-pump/Solar, 0.62–0.78 for the multimodal Nuclear/Wind/Cogen).

Table 8: Pairwise PC1, PC2, PC3 coefficients across all three reforms (functional-SA), critical hours. Cells in EUR/MWh on the curve domain (post coefficient projected via $\beta_k\phi_k(q)$, summed). The MTU15-IDA panel uses a 40-day post window. RES techs (Wind, Solar PV, Cogen, Solar Thermal) are aggregated per firm into "OTH" (RES has too many small distributed units for per-firm fPCA).

		ISP15			**MTU15-IDA**			**MTU15-DA**		
Tech	Firm	PC1	PC2	PC3	PC1	PC2	PC3	PC1	PC2	PC3
CCGT	GE	+3459	+2042	+903	+2560	-260	-1022	-15018	+879	-183
	GN	+315	+109	+32	-206	+107	-38	+862	-736	+30
	HC	-1573	+240	+175	-611	+9	-323	+2480	-270	+166
	IB	-1558	-60	-253	+16	-248	+233	+1849	-583	-218
	AXPO	-1567	-188	-139	+189	+12	+60	+1217	-609	-116
	REP	-1368	+44	-377	-258	-383	+164	+1451	-630	-163
	OTH	-1349	-61	-133	+45	-122	-90	+2183	-485	-78
Hydro	GE	+134	+124	+0	-750	-41	-22	-384	-6	+5
	GN	+351	+95	-8	+252	+58	+31	-173	-59	-42
	HC	+519	-331	+29	-243	-44	+84	+305	-264	-137
	IB	+1047	-134	-108	-269	-91	+112	-188	+25	+6
	REP	-10	-47	-20	-322	+95	+77	+55	-39	-89
	OTH	-374	+100	-8	+85	+30	+12	-702	-18	+3
Hydro pump	GE	-280	+76	-7	-359	-11	+0	-305	-16	+0
	GN	+333	+77	-9	-112	-83	+105	+1272	-19	+3
	IB	-59	-108	-41	-471	-132	+68	-57	-44	-29
	REP	-145	+1	+35	-366	-143	+89	-380	+145	-68
Nuclear	GE	-1076	-361	+64	+0	+512	+238	+1681	-767	-1498
	GN	-632	+152	+79	-65	+396	+140	+2132	-1322	+380
	IB	-842	+87	-13	+62	+393	+188	+763	-1666	-516
Wind	OTH	-50	-9	+4	+17	-1	+1	+8	+17	-12
Solar PV	OTH	+474	+39	+11	+416	-78	+33	+1599	+102	+6
Solar Thermal	OTH	+68	+11	+5	-23	+26	-26	-42	+13	+14
Cogen	OTH	-26	-9	+11	+5	-21	+14	+121	+182	-136

Reading. GE-CCGT moves opposite to GN/IB/HC under every reform — up under ISP15 (+3,459 Critical) and MTU15-IDA, sharply down under MTU15-DA (−15,018, the largest single coefficient). PC2/PC3 carry weight where a reform changes tilt rather than level: Nuclear under

MTU15-IDA has PC1 ≈ 0 but PC2 +393 to +512; CCGT under MTU15-DA has positive PC1 but negative PC2 for five of seven firms (cheap-MW tranches repriced up more than the scarcity tail). *Caveat*: the MTU15-DA coefficient bundles reforzada — only the 40-day MTU15-IDA pairwise is a clean single-reform read, and a 2022/2023 year-fold placebo flags the GE-CCGT ISP15 raw shift as seasonal.

4 Continuous-intraday repositioning per technology

The cost cascade (§8) shows expenditure shifting between real-time technical restrictions and pre-IDA Fase I; the bidding-side mechanism should be visible in the continuous intraday market (XBID). If finer-grained intraday clearing absorbs within-hour scarcity, XBID repositioning volume falls. For each (delivery date, technology) we sum XBID matched-trade MWh on the sell and buy sides — *gross* CI activity is their sum, *net* CI position is sells minus buys.

Table 9: Gross continuous-intraday activity by technology and regime, GWh/day. Sum of sell-side and buy-side MWh per (delivery date, technology), averaged per day per regime. Captures total CI repositioning by tech.

	pre-IDA	3-sess	ISP15-win	MTU15 pre-blk	MTU15 post-blk	DA15/ID15
CCGT	1.3	3.9	4.4	1.9	2.4	3.7
Nuclear	2.4	3.8	3.5	1.3	3.8	4.1
Hydro	3.3	4.2	4.5	2.6	2.4	3.2
Hydro pump	1.6	2.2	2.1	1.5	1.1	1.0
Wind	5.5	6.5	5.1	3.5	3.6	7.5
Solar PV	3.7	3.5	2.1	2.2	3.0	3.5
Solar Thermal	0.4	0.7	0.5	0.6	0.6	0.3
Biomass / RE thermal	0.4	0.6	0.5	0.5	0.4	0.4
Thermal non-RE	0.2	0.3	0.4	0.4	0.3	0.3
Battery / storage	0.0	0.0	0.0	–	0.1	0.1
Cogen	–	–	–	–	–	–

Table 10: Net continuous-intraday position by technology and regime, GWh/day. Sells − buys per (delivery date, technology). Positive values = net seller (technology adds output via CI); negative = net buyer (technology reduces output via CI).

	pre-IDA	3-sess	ISP15-win	MTU15 pre-blk	MTU15 post-blk	DA15/ID15
CCGT	+0.8	+2.5	+2.5	+1.3	+1.6	+2.2
Nuclear	−1.2	−1.0	−1.7	+0.3	−2.9	−0.2
Hydro	+0.7	+0.9	+1.1	+0.2	+0.2	+0.6
Hydro pump	+0.1	−0.0	+0.1	−0.0	+0.1	−0.1
Wind	−0.1	−0.5	−0.4	+0.5	+0.2	−1.2
Solar PV	+0.4	−0.3	−0.1	−0.3	−0.3	+0.1
Solar Thermal	−0.1	−0.1	−0.1	+0.1	−0.2	−0.1
Biomass / RE thermal	−0.1	−0.3	−0.2	−0.2	−0.2	−0.2
Thermal non-RE	−0.0	−0.0	−0.1	+0.0	−0.1	−0.1
Battery / storage	+0.0	+0.0	−0.0	–	+0.0	+0.0
Cogen	–	–	–	–	–	–

Reading.

- **System-wide gross CI activity falls during MTU15-IDA, then recovers under MTU15-DA** ($\sim$20 $\rightarrow\sim$13 $\rightarrow\sim$19 GWh/day): the granular IDA absorbs intra-hour scarcity that previously fell to XBID; the later granular DA produces more per-quarter mismatch for XBID to close.
- **Wind doubles its CI activity under MTU15-DA** (3.5 $\rightarrow$ 7.5 GWh/day, the largest tech-specific rise) — quarter-hour forecast-error correction, freed once DA and CI are both granular.

- **CCGTs are systematic net SELLERS in CI in every regime** (+0.8 to +2.5 GWh/day, Table 10): the XBID-side face of the "stay out of DA, re-enter via post-clearing channels" narrative (§5–§7). **Nuclear is the systematic net BUYER** (−1.7 to −2.9 GWh/day post-blackout), backing off as the grid favoured forced-on CCGTs.
- **Pumped storage CI activity shrinks** ($\sim 2.2 \to\sim 1.0$ GWh/day), likely migrating to the balancing-reserve markets; **Solar PV** stays roughly net-neutral (−0.3 to +0.1 GWh/day) throughout.

4.1 Seasonality-adjusted CI volumes and per-firm within-hour rebalancing

Two refinements of the volume reading. First, the per-regime CI volume deseasonalized (Fourier K=4 + DOW, Duan-smeared, the spec of §). Second, the per-firm within-hour rebalancing intensity — the standard deviation of CI sell MWh *across the four within-hour quarters* of a clock-hour — which captures offsetting within-hour adjustments (sell Q1, buy back Q3) that aggregate to near-zero hourly net but are real activity. Per-quarter resolution is essential: aggregating to hourly totals would hide it.

Table 11: Continuous-intraday daily volume by regime and hour-class, deseasonalized. GWh per day; bracketed values are Fourier+DOW adjusted (seasonality-only Spec A, Duan-smeared). The reform-window pattern — volumes fall during MTU15-IDA-only, partial recovery under DA15/ID15 — survives the SA correction in every cut.

	3-sess	ISP15-win	MTU15-IDA pre-blk	MTU15-IDA post-blk	DA15/ID15
Total system GWh/day	28.5 [32.3]	25.2 [27.0]	15.8 [12.3]	16.9 [18.1]	21.1 [22.8]
Critical hours GWh/day	14.2 [15.8]	12.3 [13.3]	7.1 [6.0]	8.3 [8.7]	10.3 [11.4]
Flat hours GWh/day	2.1 [2.6]	2.0 [2.1]	1.0 [0.8]	1.0 [1.2]	1.4 [1.6]
CCGT sell GWh/day	3.2 [3.2]	3.5 [4.0]	1.6 [2.2]	2.0 [1.9]	2.9 [2.6]
Wind sell GWh/day	3.0 [3.5]	2.3 [2.2]	2.0 [1.3]	1.9 [2.2]	3.2 [3.0]
Solar PV sell GWh/day	1.6 [1.7]	1.0 [1.4]	0.9 [0.7]	1.4 [1.2]	1.8 [2.5]
Hydro sell GWh/day	4.0 [5.3]	4.2 [3.7]	2.4 [1.8]	2.0 [2.6]	2.6 [2.4]

Reading the volume evidence. Deseasonalized total CI volume runs $32 \to 27 \to 12 \to 18 \to 23$ GWh/day across the five regimes — a −29% fall from the pre-reform baseline to DA15/ID15, with the trough (−62%) in the 40-day MTU15-IDA-only window. The fall applies to both critical and flat hours; Hydro CI activity more than halves, CCGT is roughly stable, Solar PV rises under DA15/ID15. The CI-volume decrease is the mirror image of the granular auction absorbing scarcity that previously fell to XBID and real-time TR.

Table 12: Per-firm within-hour quarter std of CI sell-side MWh, by hour-class and regime. Mean over (firm, date, clock-hour) of std(MWh_q) across the four within-hour quarters. Captures within-hour rebalancing intensity per firm. Critical hours show $\sim$ 2–5× higher within-hour variation than flat hours, consistent with granular pricing having economic content only in high within-hour residual-demand variance hours.

	Critical (MWh)			Flat (MWh)		
Firm	MTU15-IDA pre	MTU15-IDA post	DA15/ID15	MTU15-IDA pre	MTU15-IDA post	DA15/ID15
GN	9.2	8.8	**9.4**	3.0	2.1	1.7
IB	12.9	14.1	**13.0**	5.7	5.2	5.2
HC	6.5	6.8	**7.1**	3.3	4.4	4.1
GE	4.1	3.2	**2.9**	—	1.3	1.4

Reading the within-hour evidence. Within-hour quarter std in CI is $\sim$ 2–5× larger in critical than flat hours for every firm — the same critical-flat asymmetry as the auction-side bid-shape evidence (§3.4) — and ranks IB ($\sim$ 13–14 MWh) > GN ($\sim$ 9) > HC ($\sim$ 6–7) > GE ($\sim$ 3–4). It is roughly stable across the post-MTU15-IDA regimes: *CI volume fell with the* *granular auction, but per-firm within-hour rebalancing intensity did not* — firms keep making differently-sized trades across the four quarters, an activity that was invisible under hourly MTU60.

4.2 CI volumes relative to DA, IDA, and bilateral channels

Absolute CI volume can fall while its *share* of trading stays flat (if DA/IDA also fall); and the bilateral channel (PDBF − PDBC), structurally large for Spanish nuclear and hydro, may absorb activity migrating out of CI. The next table puts CI as a share of OMIE-cleared volume (DA + IDA + CI) alongside the bilateral share of dispatched volume (PDBF).

Table 13: System-wide trading-channel mix per regime, GWh/day on the regime mean. The OMIE auction-cleared volume (DA + IDA + CI) is summed across all techs each day, then averaged within regime. Bilateral = PDBF − PDBC, the contractual non-auction channel. Sample windows match Table 1.

Regime	DA	IDA	CI	Bilat	PDBF	**CI / OMIE (%)**	**Bilat / PDBF (%)**
pre-IDA	394	141	16.3	694	1 088	2.95	63.8
3-sess	460	124	21.2	628	1 088	3.50	57.7
ISP15-win	489	134	18.8	741	1 230	2.93	60.2
MTU15-IDA pre-blk	559	153	12.4	543	1 102	**1.71**	49.3
MTU15-IDA post-blk	537	151	12.7	630	1 166	1.81	54.0
DA15/ID15	400	100	14.9	711	1 112	2.89	64.0

Reading the channel mix. CI's share of OMIE-cleared volume fell from $\sim 3\%$ to $\sim 1.7\%$ during MTU15-IDA, then recovered to $\sim 2.9\%$ under DA15/ID15 — back to its pre-reform *share* even though the absolute level is lower. Bilaterals are the largest channel everywhere ($\sim 60\%$ of PDBF dispatch), dipping to 49% in the MTU15-IDA pre-blackout window and returning to 64% under DA15/ID15; per tech, renewable bilateral shares grew sharply under DA15/ID15 (Wind 20 → 43%, Solar PV 15 → 33%). The bilateral channel's reform-window growth is itself a substantive market-structure shift.

4.3 CI transaction prices vs DA prices

For each (delivery date, tech) the volume-weighted CI sell price, compared to the same-day DA spot — the *CI premium* is their difference.

Table 14: Continuous-intraday transaction prices by tech and regime, EUR/MWh. *CI premium* = CI volume-weighted price minus DA spot price for the same day. Positive premium = the technology sells in CI *above* the DA price (scarcity-side); negative = sells *below* (excess-supply-side).

	ISP15-win		MTU15-IDA post-blk		DA15/ID15			
Tech	CI px	DA px	CI px	DA px	CI px	DA px	3-sess CI prem	DA15 CI prem
CCGT	101	100	74	57	74	68	+7	+6
Hydro	101	100	71	57	71	65	+10	+6
Hydro pump	91	100	47	57	57	65	−5	−9
Nuclear	55	100	56	57	48	69	−28	−21
Wind	100	100	55	57	67	65	−3	+2
Solar PV	81	100	30	57	45	65	−21	−20

Reading. The cross-tech price structure is stable across regimes — only volumes change. CCGT and Hydro sell in CI at a small premium (+6/MWh) — the techs adding output post-DA, capturing scarcity rents; Nuclear and Solar PV sell at a large discount (−20 to −28/MWh) as excess-supply techs offloading output; Hydro pump at −5 to −9 (pump-mode arbitrage); Wind near parity.

Part II

Part B — Regulatory and confounder context

The REE program-and-intervention chain (brief primer). Every delivery day a firm program for each unit-period is built up through a chain of OMIE clearings and REE redispatch steps. Each step writes the next-level program (PDBC → PDBF → PDVP → PHF → PHFC → P48); each transition involves either a market clearing or a REE technical-restriction (RT) intervention. The schematic is:

Step	New program	Mechanism
DA clearing (D-1, 12:00)	**PDBC**	OMIE marginal-price DA cleared MW
+ bilateral contracts	**PDBF**	OMIE adds BRP-communicated bilaterals
+ REE **pre-IDA RT**	**PDVP**	*Fase I* (pay-as-bid congestion + reserve) + *Fase II* (offer redispatch)
+ IDA session s+ REE **post-IDA RT**	**PHF**(s)	Per-session IDA clear + REE security tweak ("RT2"); 3 sessions in SII
+ continuous round r+ REE post-continuous RT	**PHFC**(r)	24 rounds/day; per-round continuous + REE tweak
+ real-time RT (delivery, D)	**P48**	REE *Tiempo Real* (TR) intervention; live updates

REE's redispatch energy comes from balancing services activated alongside — FCR (primary), aFRR (secondary), mFRR (tertiary), RR, Gestión de Desvíos, RPA — which together close the PDBF-to-delivery gap; imbalance settles per ISP, dual-priced. Two recurring labels: *Reforzada* = REE's continuous post-blackout (2025-04-28 onward) stance of larger pre- and post-IDA RT, mostly Fase I-up; *RT2* = the post-IDA RT leg, measured as PHF − PIBCA at maximum-session rows. Both are confounders for the MTU15-DA comparison (its pre- and post-windows are both post-blackout). *Part B roadmap*: the phenomenon (§5, the CCGT q_1 drop), the absorbing mechanism (§6, REE post-clearing inclusion), the enabling structural feature (§7, geographic CCGT concentration), the system-cost view (§8), and the enforcement context (§9).

5 The q_1 drop: CCGT auction-cleared collapse

Variable. q_1 = sum of PDBC (OMIE DA auction-cleared) MWh, restricted to the Big-4 pivotal fleet (*Iberdrola, Endesa, Naturgy, EDP-HC*), aggregated per (technology, calendar month). Programs used: PDBC (OMIE), PDBF (PDBC + bilateral contracts, OMIE), PHF (post-IDA program at max session, OMIE). **Same-calendar-month** comparison absorbs within-year seasonality.

5.1 The three-component decomposition

The Big-4 fleet's monthly volume per technology decomposes as q_{final} = PDBC + Bilaterals + (IDA + REE-RT). The figures below separate the three components weekly for the Big-4 aggregate and for CCGT by firm: CCGT's DA-cleared layer collapses post-blackout while the IDA + RT layer balloons, and the shift is firm-wide — roughly proportional to zonal CCGT dominance (§7: GN largest, IB/GE intermediate, HC smallest) — whereas every other technology holds flat or grows on the DA layer.

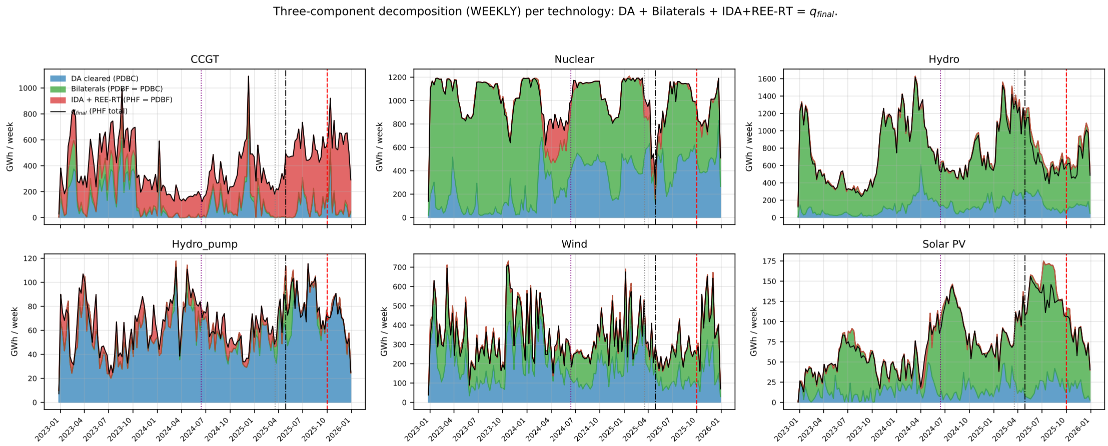


Figure 6: Weekly three-component decomposition per technology, Big-4 aggregate. PDBC (blue) + Bilaterals (green) + IDA + REE-RT (red) = q_{final} (PHF, black). CCGT's blue layer collapses post-blackout while red balloons. Vertical guides: MTU15-IDA, Blackout, MTU15-DA.

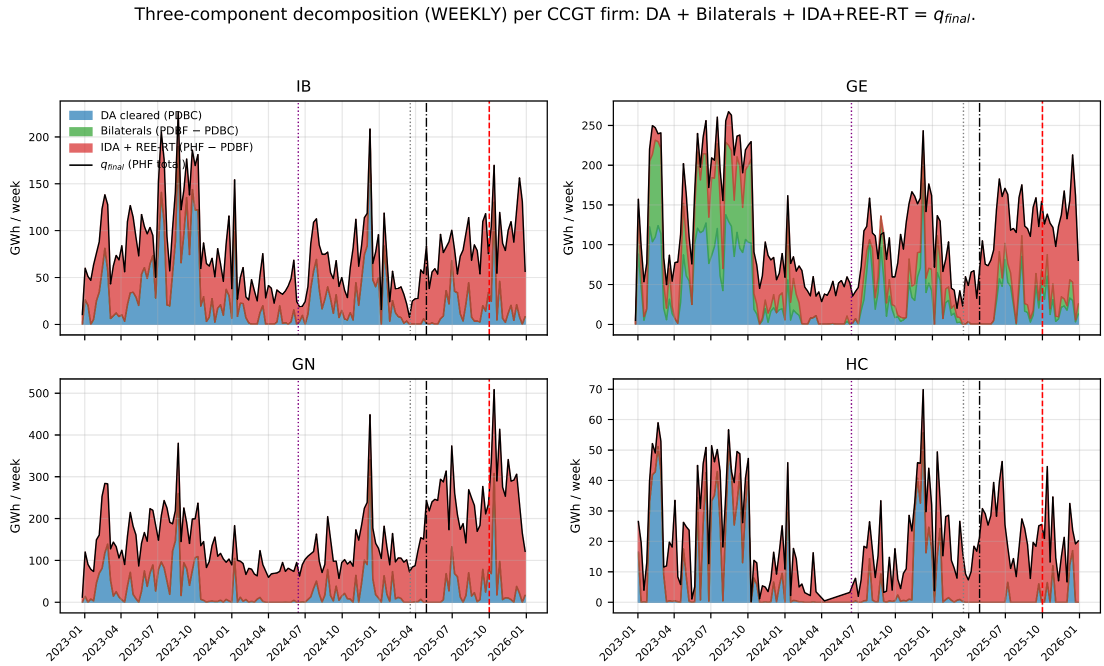


Figure 7: Weekly three-component decomposition per CCGT firm. Same construction; the DA $\rightarrow$ RT shift is firm-wide.

5.2 Headline table and reading

Table 15: Same-calendar-month auction-cleared DA volume by technology, GWh. Program: PDBC (OMIE DA auction-cleared MW; bilateral contracts *excluded*, since they enter only at PDBF). Pivotal-firm aggregate. Only CCGT collapses; every other technology grows or stays flat. *Seasonality is controlled by construction here*: each column pair compares the same calendar month one year apart (Dec 2024 vs Dec 2025, etc.), so any seasonal contribution cancels exactly. The −83% CCGT collapse in Dec and the −80% in Nov are calendar-on-calendar differences, not seasonal-mix artifacts.

	Dec 2024 → Dec 2025			Nov 2024 → Nov 2025			Aug 2024 → Aug 2025		
Tech	Pre	Post	%	Pre	Post	%	Pre	Post	%
CCGT	**1444**	**244**	**-83%**	**637**	**130**	**-80%**	**532**	**360**	**-32%**
Nuclear	1253	2587	+106%	933	1909	+105%	1997	2243	+12%
Hydro	1016	640	-37%	509	611	+20%	303	424	+40%
Hydro_pump	195	247	+27%	182	341	+87%	239	406	+70%
Wind	732	696	-5%	666	1081	+62%	379	449	+18%
Solar PV	57	34	-40%	37	100	+170%	37	136	+268%

Table 16: q_1 per-firm, per-hour-class same-calendar-month decomposition (mean DA-cleared MWh per unit-period-cell). Disaggregation of Table 15 to (tech, firm, hour-class). Same Dec24/Dec25, Nov24/Nov25, Aug24/Aug25 month pairs. CCGT critical-hour drops are uniform across firms (IB −11%, GE −6%, GN −10%, HC −18% Dec24→25); CCGT flat-hour drops are larger (GN −30%, HC −50%). Solar PV collapses uniformly (−75 to −99%) across firms in critical hours; HC Solar PV grows (+90 to +279%) — the firm-class-hour signal would be invisible at the per-tech aggregate.

			Dec 24 -¿ Dec 25			Nov 24 -¿ Nov 25			Aug 24 -¿ Aug 25		
Tech	Firm	Hour-cl	Pre	Post	%	Pre	Post	%	Pre	Post	%
CCGT	IB	Critical	+368	+329	-11%	+314	+327	+4%	+316	+276	-13%
		Flat	—	—	—	+125	+299	+140%	+197	+210	+6%
	GE	Critical	+383	+360	-6%	+343	+349	+2%	+304	+349	+15%
		Flat	+324	+375	+16%	—	—	—	+311	+350	+13%
	GN	Critical	+354	+318	-10%	+345	+309	-11%	+314	+291	-7%
		Flat	+323	+225	-30%	+386	+326	-15%	+353	+352	-0%
	HC	Critical	+359	+295	-18%	—	—	—	—	—	—
		Flat	+308	+153	-50%	—	—	—	—	—	—
Hydro	IB	Critical	+674	+96	-86%	+338	+174	-49%	+219	+201	-8%
		Flat	+352	+29	-92%	+297	+75	-75%	+139	+146	+5%
	GE	Critical	+100	+117	+17%	+106	+114	+7%	+58	+77	+35%
		Flat	+72	+110	+53%	+104	+116	+12%	+54	+93	+71%
	GN	Critical	+145	+117	-19%	+114	+62	-45%	+97	+97	+0%
		Flat	+156	+44	-72%	—	—	—	—	—	—
	HC	Critical	+11	+6	-50%	+5	+25	+401%	+4	+22	+442%
		Flat	+1	+1	+9%	—	—	—	+1	+21	+2976%
Hydro pump	IB	Critical	+662	+769	+16%	+569	+934	+64%	+861	+996	+16%
		Flat	—	—	—	—	—	—	+594	+785	+32%
	GE	Critical	+77	+99	+29%	+95	+101	+7%	+82	+118	+42%
		Flat	+36	+49	+36%	+49	+73	+49%	+60	+96	+61%
Nuclear	IB	Critical	+92	+348	+281%	+60	+302	+401%	+35	+143	+307%
		Flat	+99	+374	+276%	+68	+359	+429%	+21	+110	+431%
	GE	Critical	+792	+893	+13%	+794	+912	+15%	+846	+845	-0%
		Flat	+793	+900	+13%	+806	+917	+14%	+847	+849	+0%
Wind	IB	Critical	+518	+32	-94%	+245	+83	-66%	+128	+31	-76%
		Flat	+573	+87	-85%	+347	+128	-63%	+135	+10	-93%
	GN	Critical	+26	+35	+36%	+31	+38	+23%	+16	+18	+11%
		Flat	+31	+43	+40%	+36	+50	+39%	+19	+18	-6%
	HC	Critical	+133	+132	-1%	+132	+187	+42%	+106	+114	+8%
		Flat	+130	+126	-3%	+136	+191	+41%	+121	+123	+1%
Solar PV	IB	Critical	+13	+3	-75%	+9	+11	+26%	+49	+29	-41%
	GE	Critical	+1	+0	-89%	+1	+0	-92%	+0	+0	-64%
	GN	Critical	+13	+0	-99%	+3	+1	-59%	+2	+0	-88%
	HC	Critical	+6	+11	+90%	+5	+15	+204%	+14	+51	+279%

Reading. (i) CCGT q_1 Big-4 fleet: Dec 2024 = 1 444 GWh → Dec 2025 = 244 GWh (−83%). Nov same comparison: −80%. (ii) Nuclear, Wind, Hydro_pump, Solar PV *grow* in the same window. (iii) The intensive margin (per-cell mean MWh *conditional* on being cleared) is stable for CCGT at ≈300 MWh; what collapses is the extensive margin (number of unit-day-hours cleared at all). (iv) Only CCGT has the strategic option to stay out of DA and rejoin via post-clearing redispatch (mainly pre-IDA Fase I) and the intraday auctions: renewables are not called up in PO 3.2 (their increase offers are primary-energy-constrained) and nuclear is baseload-must-clear. (v) The q_1 drop is broadly uniform within CCGT firms across hour-classes (Table 16); Solar PV collapses uniformly across IB/GE/GN while HC Solar PV grows.

Verdict. The q_1 drop is mechanism-specific: CCGT firms with strong post-clearing backstop (zonal dominance, §7) increasingly bid above DA clearing under *operación reforzada*, are rescued by REE through pay-as-bid Fase I (§6), and post a near-constant cap-padded DA bid ladder (§7.1). The collapse is the empirical face of the practice CNMC sanctioned in 2019 (§9), now at fleet-wide scale.

5.3 Parallel-trends on the post-clearing gap

Table 17: Post-DA gap by technology, mean GWh / month. Programs: PHF − PDBF. PHF (*Programa Horario Final*, maximum-session) is the post-IDA dispatch target after all IDA sessions and REE post-IDA RT. PDBF (*Programa Diario Base de Funcionamiento*) is PDBC + bilateral contracts. Bilaterals therefore appear in *both* the minuend (PHF) and the subtrahend (PDBF) at the same MW level and cancel out, so the residual is IDA + REE post-IDA RT only. The bilateral channel itself (PDBF − PDBC) is reported separately in Table 18.

Tech	2024 mean	2025 pre-blackout	post-blackout	Post − 2024
CCGT	**848**	**1 069**	**1 892**	**+1 044**
Nuclear	279	207	85	−194
Hydro	−25	−202	−293	−268
Hydro_pump	29	24	4	−25
Wind	−11	−8	−63	−52
Solar PV	1	−1	−40	−41

Reading. (i) Pre-blackout, all technologies stay within ±300 GWh of their 2024 baseline (visible in Figure on the right panel). (ii) CCGT pre-drift +220 GWh from 2024 to 2025 pre-blackout, modest. (iii) Post-blackout, CCGT alone breaks parallel trends with +1 044 GWh jump — ≈ 5× the pre-drift, a structural break not a continuation of trend. (iv) Nuclear / Hydro / Wind / Solar all *lose* volume on the same channel (−25 to −268 GWh): REE displaces clean tech post-DA to make room for the forced-on CCGTs, consistent with the per-firm decomposition in §6.

Seasonality robustness. The CCGT post-blackout jump survives Fourier-deseasonalization. Running $\text{gap}_t = \alpha + \sum_r \beta_r \mathbf{1}\{\text{regime}_t = r\} + \sum_{k=1}^{4} \text{Fourier}_k(t) + \varepsilon_t$ on the daily CCGT post-DA gap series (PHF-PDBC, all CCGT units) pooled 2022-2026 gives a regime-mean CCGT gap that rises from ∼47 GWh/day (3-sess baseline) to ∼389 GWh/day (MTU15-IDA post-blackout) to ∼417 GWh/day (DA15/ID15) at *average day-of-year*. The structural break is preserved.

5.4 Bilateral channel as a sanity check

The q_1 collapse uses PDBC (auction-cleared only). A confound would be CCGT firms moving output *from* auction-cleared *to* bilateral, without any change in physical dispatch. Table 18 shows the bilateral channel directly. For CCGT the channel is small (∼ 90–128 GWh/month, no upward trend across the reform window); the auction-cleared collapse is real, not a relabel of output as bilateral. For other technologies (Hydro, Nuclear, Wind) the bilateral channel is structurally large — 1–4 TWh/month — and dominates the auction-cleared volume; their PDBC numbers in Table 15 therefore understate physical output by a large constant offset, but the *year-over-year change* is still informative.

Table 18: Bilateral channel by technology and regime, GWh / month. Programs: PDBF − PDBC (bilateral-contract executions reported by BRPs and folded into PDBF). Pivotal-firm aggregate. CCGT bilaterals are small and roughly stable; Hydro/Nuclear/Wind bilaterals are an order of magnitude larger and structurally part of their dispatch.

	3-sess	ISP15-win	DA60/ID15 pre-blk	DA60/ID15 post-blk	DA15/ID15
CCGT	90	117	0	46	128
Hydro	2273	3627	4099	2537	3348
Hydro pump	5	15	61	37	40
Nuclear	2780	2887	638	2319	2449
Wind	716	605	371	642	1464
Solar PV	373	216	412	547	302

5.5 Identification implication

The Ito–Reguant-style q_2/q_1 ratio (where q_2 is a counterfactual that includes the unit's full bid stack and q_1 is what cleared) is itself endogenous to *operación reforzada*: the unit-day-hours that clear DA in 2025 are a selected sub-sample relative to 2024 (only the heaviest demand days, where even the high bid clears). Same-calendar-month + unit FE help but do not fully address this composition shift. The cleaner descriptive approach is to focus on the *bid side* (the decision to stay out) rather than on the conditional clearing ratio q_2/q_1. The bid-side reading is in §7.1.

6 REE post-clearing inclusion as the absorbing mechanism

6.1 Three measurement paths

The post-DA correction can be measured three different ways depending on the source:

- **Path A** — per-unit **PHF − PDBF** from OMIE programs (max-session PHF). Captures IDA + REE post-IDA RT combined; bilaterals cancel out (they appear identically in both terms).
- **Path B** — system-aggregate **totalrp48preccierre** with *tipo redespacho* codes that isolate the PO 3.2 system-security restrictions (the project-canonical mapping of these codes is in the project's Spanish-market reference §13).
- **Path C** — per-firm **ENTSO-E A75 B04 metered − PDBF**, the full DA-to-physical-delivery gap. Coarser per firm because A75 lacks unit-level resolution.

Path A is the workhorse measurement throughout this section. Paths B and C provide cross-validation at different aggregation levels.

6.2 Full post-clearing intervention by *tipo redespacho* code

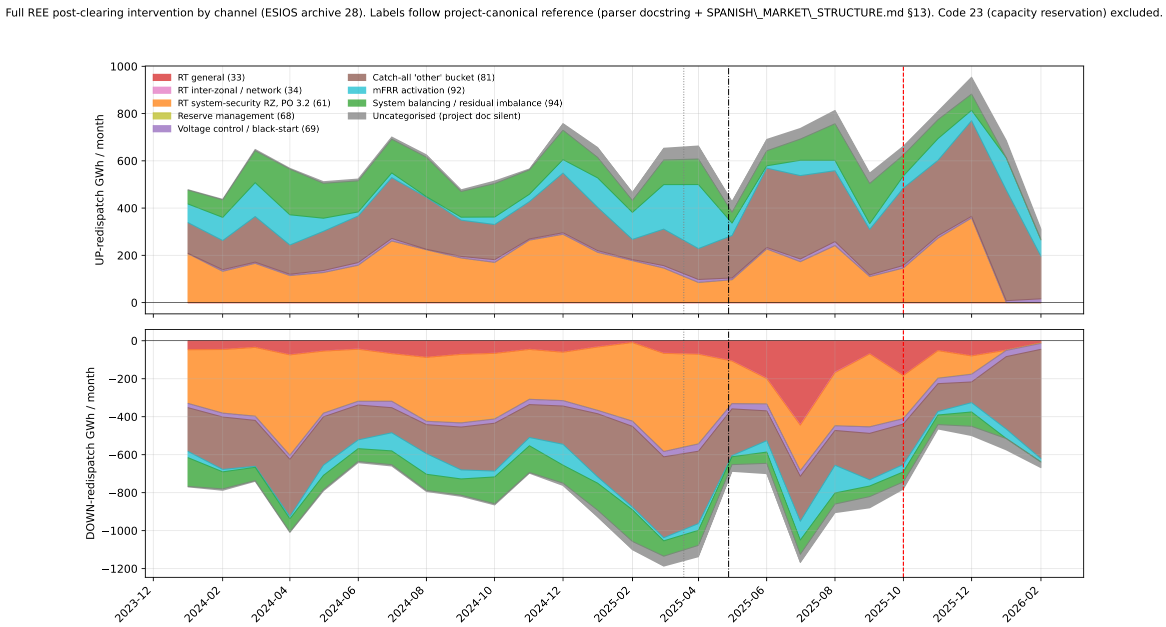


Figure 8: Full REE post-clearing intervention by *tipo redespacho* code, ESIOS archive 28 (`totalrp48preccierre`). Top: UP-redispatch GWh / month; bottom: DOWN-redispatch (negative axis). Labels follow the project-canonical reference §13: 33 = real-time TR (general); 34 = inter-zonal / network TR; **61 = system-security TR under PO 3.2 (the post-IDA / real-time channel)**; 68 = reserve management; 69 = voltage control / black-start; 81 = catch-all "other"; 92 = mFRR activation; 94 = system balancing. Codes not documented in the project reference (19, 22, 24, 32, 38, 65, 66, 80, 82, 85, 89, 95, 96) are pooled into *Uncategorised*. Code 23 ($\sim$10.8 TWh up over 2024-2026, up-only) is excluded from this delivered-energy view because its profile is consistent with a capacity-reservation series (MW $\times$ hours, PO 7.2 banda-secundaria); stacking it on the same axis as activated-energy channels would be apples-to-oranges.

6.3 Per-firm $\times$ per-technology Path A reading

Programs: PHF − PDBF, per (firm, technology, month). Pre window 2024-01 $\to$ 2025-03; post window 2025-05 $\to$ 2026-01 (*operación reforzada*). Bilaterals cancel out (in both terms at the same MW); the residual is IDA + REE post-IDA RT.

Table 19: Post-DA intervention by firm and technology, GWh / month. Programs: PHF − PDBF, same construction as Table 17 but resolved per firm. The bilateral channel (PDBF − PDBC) is small for CCGT (Table 18); does *not* drive this gap.

Tech	Firm	Pre (GWh/mo)	Post (GWh/mo)	Δ (GWh/mo)	Ratio
CCGT	GE	213	413	+200	1.94
	GN	397	932	+534	2.34
	HC	42	94	+52	2.25
	IB	146	316	+169	2.16
Hydro	GE	-58	-253	-195	4.33
	GN	5	-33	-38	-6.96
	HC	-0	-11	-11	40.05
	IB	33	-84	-117	-2.55
Hydro_pump	GE	2	-15	-17	-7.14
	GN	4	3	-1	0.70
	IB	23	-1	-24	-0.03
Nuclear	GE	241	23	-218	0.10
	IB	1	-137	-138	–
Wind	GN	-9	-36	-27	4.02
	HC	-13	-50	-36	3.71
	IB	8	-81	-89	-10.63
Solar PV	GE	-0	-0	-0	9.13
	GN	-3	-11	-8	3.29
	HC	0	1	+1	6.26
	IB	3	-42	-45	-12.10

Reading. (i) Every CCGT firm doubles its post-DA gap; GN gains the most (+534 GWh/mo). (ii) The mirror decrease in Wind / Solar / Hydro / Nuclear shows that REE substitutes CCGT for clean tech post-PDBF. (iii) Cross-checking against the system view in Figure 8: the two largest delivered-energy channels in 2024-2026 are the PO 3.2 system-security restrictions (code 61, ∼12 TWh each direction) and the catch-all code 81 (∼12 TWh each direction; the latter's exact content is not documented in the project reference and we treat it as not interpretable at the channel level). Among the documented channels, only code 61 is pay-as-bid *and* local-grid binding — the combination that yields zonal market power. (iv) Below the hour-class mean, critical hours carry ∼ 3–5× the per-day gap of flat or midday for every CCGT firm, and the reform-window shift is whole-distribution rather than tail-driven.

Verdict. Pay-as-bid pricing of RT-forced volume × zonal dominance (§7) is a local-monopoly channel. The aFRR / mFRR / GD / RR channels in Figure 8 are reported for completeness but they are competitively priced (system-marginal clearing on PICASSO / MARI / TERRE) and not subject to local-zone bargaining. Only PO 3.2 system-security TR (code 61) carries the pay-as-bid + local-grid binding combination.

6.4 Per-firm CCGT post-clearing revenue (proxy)

Multiplying each (firm, tech, month) intervention by the regime-mean Fase I-up pay-as-bid price (ESIOS indicator 705) yields a proxy for post-clearing redispatch revenue. The proxy sums Fase I + Fase II + TR up redispatch revenues (cannot isolate Fase I alone without unit-level pairing). It under-attributes RES-curtailment compensation, which uses a separate price series. Big-4 CCGT only:

Table 20: Per-firm CCGT up-redispatch revenue (proxy), EUR million per regime window. Computed as $\sum_{m \in \text{regime}}$ (intervention MWh)$_m$ $\times$ regime-mean Fase I-up pay-as-bid price (ESIOS indicator 705). The Big-4 sum in DA15/ID15 ($\sim$ 1,390M) sits close to the REE-published Fase I-up cost (1,200M in Table 26); the $\sim$ 15% overshoot reflects the inclusion of TR-up and Fase II-up volumes that our PHF $-$ PDBF gap aggregates into the same Fase I-up multiplier.

Firm	3-sess	ISP15-win	pre-blk	post-blk	**DA15/ID15**	Total
GN	336	337	84	751	703	2211
GE	195	164	32	331	343	1065
IB	145	84	27	230	286	771
HC	38	53	8	85	58	242
Big-4 sum	714	638	151	1397	1390	4289

Reading. **GN captures $\sim$ 50% of Big-4 CCGT post-clearing revenue** in every regime. The reform-window evolution for GN CCGT: €336M (3-sess) $\rightarrow$ €337M (ISP15-win) $\rightarrow$ €84M (40-day pre-blk, ~€766M annualised) $\rightarrow$ €751M (post-blk) $\rightarrow$ €703M (DA15/ID15). In per-day terms: ~€2M/day pre-reforzada, ~€5–8M/day post-reforzada. **The reform-induced increase in CCGT post-clearing revenue is geographically concentrated** in GN's southern-corridor zones (Sur, Levante, Cataluña per Table 22). The operationally-required-by-REE CCGTs are the ones that capture the Fase I rent; firms outside the voltage-fragile zones (HC, smaller share of IB and GE) capture less.

6.5 CCGT marginal cost and the Fase I scarcity rent

To assess how much of the Fase I price represents rent above short-run marginal cost (SRMC), we compute a transparent bottom-up MC estimate for the Spanish CCGT fleet using observable fuel and CO_2 inputs.

Formula.

$$\mathrm{MC}_{\mathrm{CCGT}} = \frac{P_{\mathrm{gas}}}{\eta} + \frac{P_{\mathrm{CO_2}} \cdot \mathrm{EF}}{\eta} + \mathrm{VOM}$$

where P_{gas} is the natural-gas spot price (EUR/MWh-thermal, LHV basis), η is the thermal-to-electric efficiency, $P_{\mathrm{CO_2}}$ is the EUA spot price, EF is the gas combustion emission factor (tCO_2/MWh-thermal), and VOM is variable operations & maintenance.

Parameters. $\eta = 0.55$ LHV (Spanish fleet, 2002–2008 vintage, 52–57% range; IEA *Projected Costs* 2020) — the central estimate, with $\eta = 0.52$ as a conservative lower bound on the premium; EF = 0.20 tCO_2/MWh-thermal (IPCC 2006 default for natural gas); VOM = 4 EUR/MWh-elec (upper end of the IEA range, conservative). P_{gas} from ESIOS id 1940 (TNP hub, converted at 13.33 MWh/ton); $P_{\mathrm{CO_2}}$ from ESIOS id 1391. Start-up/no-load costs are excluded from the formula and add $\sim$ 5–15 EUR/MWh-elec for units called specifically for Fase I, so the headline premium below overstates true rent by $\sim$ 10 EUR/MWh. ESIOS id 705 (Fase I-up) is the accepted-energy pay-as-bid price, not a capacity reservation price; the opportunity-cost benchmark is the unit's SRMC.

Result. Monthly aggregates with the central parameters give CCGT marginal cost in the range €100–130/MWh-elec across the reform-window months, against DA spot €53–120/MWh and Fase I-up pay-as-bid €145–261/MWh (Table 24):

Table 21: Approximate CCGT marginal cost vs market and Fase I prices, per regime. All in EUR/MWh-elec. $MC_{central}$ from monthly ESIOS gas + CO_2 prices using $\eta = 0.55$, EF = 0.20, VOM = 4. "Premium over MC" is the Fase I up price minus $MC_{central}$ *before* adding the $\sim$ 5–15 EUR/MWh start-up/no-load adjustment discussed above. After start-up adjustment, the true rent is roughly $\sim$ 10 EUR/MWh smaller.

Regime	DA spot	Fase I up (id 705)	$MC_{central}$	Premium over MC
3-sess (Jun–Nov 24)	77	159	$\sim$ 100–115	+44 to +59
ISP15-win (Dec24–Mar25)	118	187	$\sim$ 125–150	+37 to +62
DA60/ID15 pre-blk	54	145	$\sim$ 130	+15
DA60/ID15 post-blk	63	165	$\sim$ 100–110	+55 to +65
DA15/ID15	76	168	$\sim$ 100–110	+58 **to** +68

Reading. The Fase I-up price exceeds bottom-up CCGT marginal cost by 40–70 EUR/MWh in every regime, i.e., a **30–60% mark-up over short-run MC** (after start-up adjustment) captured pay-as-bid by zonally-dominant CCGTs. DA spot in 2024–2026 sits at or below CCGT MC much of the time.

Caveats. Reading prices against MC is cleaner in Fase I (pay-as-bid: any accepted bid above MC is positive cash flow) than in DA, where a bid above MC could be either competitive non-clearing or strategic markup. And the premium is scoped to the units that clear Fase I — disproportionately zonally-constrained-corridor CCGTs ($\sim$ 20–30% of the fleet, Table 22), not "Spanish CCGTs" in general.

7 Geographic CCGT concentration and zonal market power

Why this matters. PO 3.2 redispatches relieve transmission-grid constraints *locally*, and the resulting redispatch energy is paid pay-as-bid (§6). The zonal-dominant CCGT operator therefore holds local market power on the Fase I / TR bids in its zone, independent of the EUPHEMIA day-ahead clearing. The fewer the eligible CCGTs in a constrained zone, the higher the resulting Fase I price.

Variable. CCGT installed capacity (MW) per (zone, firm), with HHI computed on capacity shares within each zone (×10,000). **Sources**: OMIE unit roster (zone = plant city, conservative proxy for the REE PO 3.2 zonal book) joined with ESIOS archive 110 (master data on registered installations).

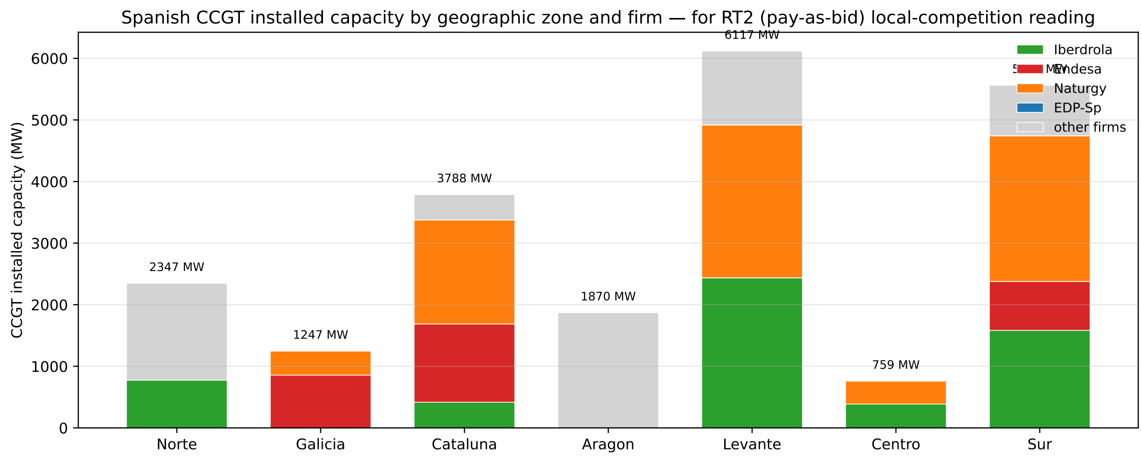


Figure 9: Spanish CCGT installed capacity (MW) by zone and firm. Bars stack each firm's MW; zone labels (Sur, Levante, etc.) are grouped from plant-city assignments. Galicia and Centro are duopolies; Sur / Levante / Cataluña have wider firm diversity but GN holds the largest share in each.

Table 22: Per-zone CCGT capacity concentration. HHI on within-zone capacity shares (×10,000).

Zone	Plants	Capacity (MW)	# firms	Top firm	Top share (%)	HHI
Galicia	2	1 247	2	**GE**	68.6	5 695
Norte	10	2 347	2*	non-Big-4*	67.0	5 577
Aragón	3	1 870	2*	non-Big-4*	57.7	5 119
Centro	2	759	2	**IB**	50.9	5 001
Levante	12	6 117	3	**GN**	40.6	3 616
Cataluña	8	3 788	4	**GN**	44.5	3 348
Sur	13	5 563	4	**GN**	42.5	3 040

* Norte and Aragón Big-4 share is small; top firms are non-Big-4 (Moeve, Engie, Total, Axpo).

Reading. (i) **Galicia** (HHI 5,695): 2 CCGT plants only, GE at 69%. (ii) **Centro** (HHI 5,001): 2 plants, IB 51% / GN 49%. (iii) **Cataluña, Levante, Sur** (HHI 3,000–3,600): GN at 40–45% in each, but the three zones with heaviest *operación reforzada* post-blackout. (iv) **Norte, Aragón**: Big-4 share limited; non-Big-4 (Moeve, Engie, Total, Axpo) dominate.

Verdict. GN's zonal dominance in the southern-Mediterranean corridor (Sur, Levante, Cataluña, plus Centro through IB) maps directly onto the post-clearing intervention story in §6. Pay-as-bid Fase I / TR + zonal dominance is a local-monopoly channel that does not require flow-through to the DA auction. *Caveats*: zone = plant city (a conservative proxy for the REE PO 3.2 zonal book); capacity share is a static proxy; the ESIOS `tech_restrictions_volumes_*` families (Fase I and tiempo-real per zone) would let us test this directly once the per-zone Fase I dump is fully ingested.

7.1 Bid-side heterogeneity by zonal dominance

The capacity-based zonal map (Table 22) says *where* firms could exercise local market power; the question this subsection answers is *whether they bid differently when they hold dominance versus when they do not.* We split each CCGT unit into "dominant in its zone" (the firm holds the largest within-zone capacity share) versus "non-dominant" and compare the median day-ahead sell-bid price per firm per regime.

Variable. For each (CCGT unit, day) the median day-ahead sell-bid price across the day's quarters (EUR/MWh). Aggregated by (firm, regime, dominant flag) with the median across (unit, day) cells. **Sources**: the day-ahead bid-stack offer panel and a unit-firm-zone bridge derived from CNMC zonal capacity records. "Pre" = 2024 same-calendar; "post" = post-blackout window 2025-05 onward.

Table 23: Median DA sell-bid price by firm and zonal dominance, EUR/MWh. Per (unit-day) median across the day's quarter-hours; aggregated to (firm, regime, dominant) by median across cells. "Dominant" = firm holds the largest CCGT capacity share in the zone. *Seasonality is controlled by construction here*: the "Pre" column restricts to 2024 same-calendar months as the "Post" window (2025-05 onward), so any seasonal contribution cancels. The GE dominant-vs-non-dominant differential (+1,344 post, ratio 4×) and the GN uniform-cap-ladder pattern are calendar-matched, not seasonal-mix artifacts.

	Pre (2024 same-cal)				Post (2025-05+ reforzada)			
Firm	Dom	Non-dom	Diff	ratio	Dom	Non-dom	Diff	ratio
GN	844	866	−22	0.97	840	852	−12	0.99
GE	**2,680**	2,381	+299	1.13	**1,789**	445	+1,344	**4.02**
IB	156	170	−14	0.92	147	196	−49	0.75
HC	–	156	–	–	–	163	–	–

GN unit-counts: 15 dominant (Sur/Levante/Cataluña), 2 non-dominant. GE unit-counts: 1 dominant (Galicia: PGR5), 4 non-dominant (Cataluña: BES3/BES5; Sur: COL4/SROQ2). IB: 1 dominant (Centro), 9 non-dominant. HC has no dominant-zone CCGT in this taxonomy.

Reading. Three distinct bidding strategies surface across the four firms:

- **GN: cap-padded ladder applied uniformly.** GN bids at €840–870/MWh in *every* zone, dominant or not, in both regimes. The bid level barely shifts post-blackout. This is the empirical signature of a fleet-wide Cournot-on-quantity strategy applied as company policy — not a per-zone tactical adjustment — consistent with the Cournot reading in §9.1.
- **GE: zone-specific bid level, sharply asymmetric.** GE bids €2,680/MWh in Galicia (its dominant zone, with PGR5 the sole flagship unit) pre-reform and €1,789/MWh post; in non-dominant zones, the same firm bids €2,381 pre and collapses to €445 post — a −81% drop, 4× ratio between dominant and non-dominant zones in 2025. The underlying unit-level detail confirms this: PGR5 in Galicia holds €1,789 in 2025; BES3 in Cataluña drops from €1,346 to €473; COL4 in Sur drops from €2,476 to €109. GE concentrates the deep-withholding posture on the units where its zonal-Fase I outside option is largest, and lets the non-dominant-zone units clear DA at low prices.
- **IB and HC: bid near short-run marginal cost everywhere.** IB bids €150–220/MWh and HC €150–165, both close to bottom-up CCGT marginal cost (§6.5). IB has only one dominant-zone unit (Centro) and shows no dominant-vs-non-dominant differential. HC has no dominant-zone CCGT.

Verdict. Per-zone bidding behaviour confirms the local-market-power reading rather than refuting it. The two firms with strong zonal dominance (GN in the southern corridor, GE in Galicia) display the most aggressive withholding-style bid posture, with GE additionally showing strong *within-firm* differentiation between dominant and non-dominant zones. The two firms with thin or no dominant footprint (IB outside Centro, HC) bid near competitive levels everywhere. This complements the capacity-based concentration evidence (Table 22) with a bid-side observation: dominance is exercised. The CNMC's 2019 Naturgy sanction precedent (§9.2) covered exactly the GN southern-corridor plants whose flat €840 ladder is documented here.

8 System cost cascade: the full ajuste view

Every MWh cleared in DA/IDA triggers downstream settlements in ≥ 6 ancillary processes whose combined cost can rival the DA energy bill in expensive periods. If the reform shifts cost rather than removing it, total system cost would not fall — the cascade is the falsifier.

Service inventory. REE's adjustment markets, in delivery-time order: *Restricciones Fase I* (pre-IDA, PO 3.2, **pay-as-bid**); *Fase II* (post-IDA, operationally dead post-2024); *Gestión de Desvíos* (PO 7.3); *Restricciones Tiempo Real* (TR, post-gate-closure, PO 3.2, **pay-as-bid**); *banda secundaria* (aFRR capacity, daily auction, marginal); *aFRR energy* (PICASSO), *mFRR* (MARI), *RR* (TERRE), all marginal-priced per cycle; *SRAD* (demand response, small); *Desvíos* (imbalance settlement, PO 14.4, dual-price). FCR is mandatory-unpaid; voltage control (PO 7.4) became a paid product mid-2025. The pay-as-bid Fase I and TR legs are where local market power can bite; the marginal-priced balancing services do not.

8.1 Per-regime price comparison, by direction

For each priced service, the mean accepted price (EUR/MWh) across ISP cells per regime — DA spot from OMIE, the ajuste legs from the corresponding ESIOS price indicators.

Table 24: Per-regime average UP price by service, EUR/MWh. "—" = no data in window. Sources: ESIOS indicator IDs cited inline in the section preamble; arithmetic mean across ISP-15 cells. *Bracketed (SA) values*: Duan-smeared Fourier+DOW seasonally adjusted predictions, applied to outcomes with multi-year daily series (DA spot from OMIE; aFRR reserve from ESIOS id 634). DA spot decline strengthens post-deseasonalization (ISP15-win 132.7 (SA) → DA15/ID15 65.3 (SA), winter raw inflated by heating-demand seasonality). aFRR reserve same direction. Other rows kept raw.

	3-sess	DA15/ID15	DA60/ID15 post-blk	DA60/ID15 pre-blk	ISP15-win
DA spot	76.7 [74.4]	76.4 [65.3]	63.2 [50.2]	54.0 [54.0]	117.5 [132.7]
Fase I (PDBF) up/dn	158.5	168.3	165.3	145.1	187.2
Fase II up/dn	92.1	20.9	43.4	2.1	143.0
Imbalance cobro/pago	48.6	51.3	39.0	4.3	69.4
RR (rrenergyprice)	30.4	41.5	-10.6	5.7	89.5
TR (Tiempo Real) up/dn	451.3	226.0	220.1	226.4	545.0
aFRR energy	82.0	75.6	69.0	42.8	104.0
aFRR reserve (cap.)	14.7 [9.9]	6.5 [3.5]	6.3 [4.7]	5.5 [4.8]	13.3 [7.3]
mFRR programada	99.9	—	—	—	131.3

Table 25: Per-regime average DOWN price by service, EUR/MWh. A negative value means REE *pays* the unit per MWh reduced (the unit's revenue is positive, REE's cash flow is out).

	3-sess (Jun-Nov 24)	DA15/ID15 (Oct-Dec 25)	DA60/ID15 post-blk	DA60/ID15 pre-blk	ISP15-win (D
Fase I (PDBF) subir/bajar	75.8	1779.5	171.5	48.1	
Fase II subir/bajar	76.5	67.2	56.5	22.1	
Imbalance cobro/pago	91.9	80.9	67.7	33.1	
Imbalance medido up/dn	91.9	80.9	67.7	33.1	
TR (Tiempo Real) subir/bajar	-8.1	-30.7	-63.0	-102.6	
aFRR energy	51.3	36.7	30.0	6.1	
aFRR reserve (capacity)	7.9	4.1	4.3	6.3	
mFRR programada	35.3	—	—	—	

8.2 Per-regime cost composition (REE-published cost indicators)

Variable. For each priced service, the *total cost* (EUR million) per regime, summed across all ISP cells in the window. **Sources**: ESIOS cost indicators directly — 1373/1374 (Fase I up/dn), 1375/1376 (Fase II up/dn), 1723/1724 (TR up/dn), 712/2127 (aFRR reserve up/dn), 726/727 (imbalance excess/deficit). We exclude id 899/2129 (aFRR assignment) since it duplicates id 712/2127 (the same flow viewed from another angle); we exclude id 709/724 (daily totals) since they aggregate the per-direction breakdowns.

Table 26: Per-regime ajuste-cascade cost composition. EUR *million* per regime, summed across ISP-15 cells. **Source**: ESIOS cost indicators (REE-published, per-day or per-ISP, summed). Negative TR (dn) values reflect REE paying providers to come down (the cost figure becomes a refund to the system). *Bracketed (SA) values for Fase I (up) and TR (up)*: Duan-smeared Fourier+DOW seasonally adjusted per-day rate × regime days. Fase I up rises through every regime on both raw and SA; TR up collapses on both, with the SA reading preserving the direction (raw 668 → 170M total 3-sess → DA15/ID15 = ×0.25; SA ∼ 611 → 273M total = ×0.45 — both confirm structural collapse, the raw understates it). Other rows kept raw.

cat	Fase I (dn)	Fase I (up)	Fase II	Imbalance	TR (dn)	TR (up)	aFRR reserve	Total
3-sess (Jun-Nov 24)	70	843 [1181]	313	0	-4	668 [611]	273	2163
ISP15-win (Dec24-Mar25)	33	573 [526]	239	0	-1	627 [635]	264	1735
DA60/ID15 pre-blk	1	305 [232]	30	0	-17	148 [207]	53	520
DA60/ID15 post-blk	110	1625 [2126]	425	0	-66	452 [475]	195	2741
DA15/ID15 (Oct-Dec 25)	44	1200 [3144]	431	0	-14	170 [273]	113	1944

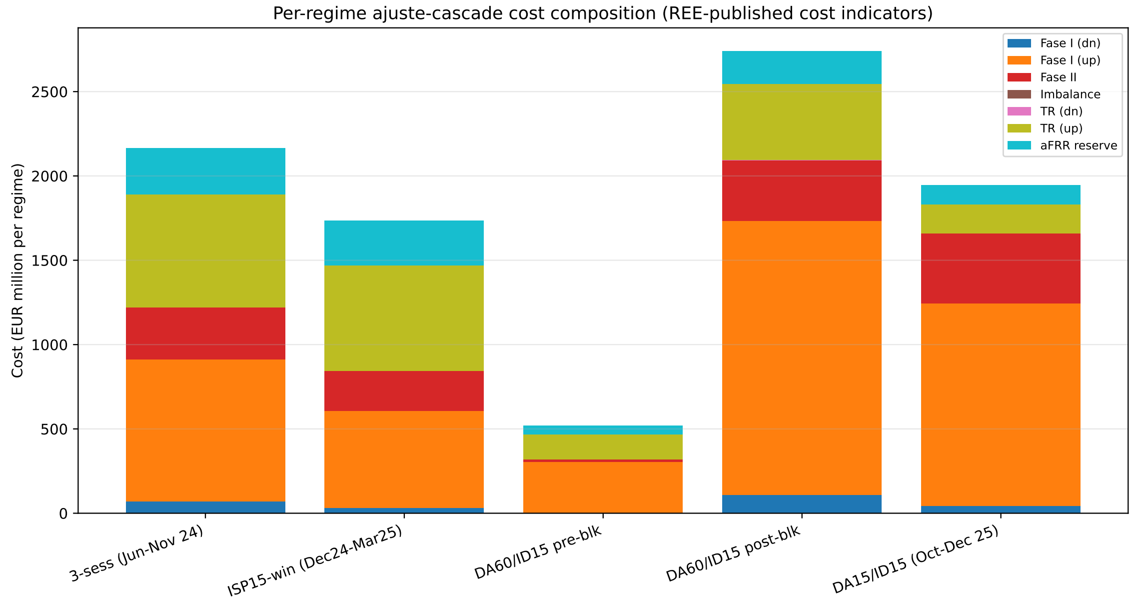


Figure 10: Stacked cost composition per regime. Bar height = total ajuste cost. Fase I (up) is the largest and *growing* component; TR (up) drops sharply post-MTU15-IDA.

8.3 Per-day equivalents (the headline)

	3-sess	ISP15-win	DA60/ID15 pre-blk	DA60/ID15 post-blk	**DA15/ID15**
Length (days)	∼170	∼108	40	∼155	92
Total cost (€M)	2 163	1 735	520	2 741	1 944
Per day (€M)	12.7	**16.1**	13.0	**17.7**	**21.1**
Fase I up / day	5.0 [7.0]	5.3 [4.9]	7.6 [5.8]	10.5 [13.6]	**13.0** [13.9]
TR up / day	3.9 [3.6]	5.8 [5.9]	3.7 [5.2]	2.9 [3.0]	1.8 [1.2]

[bracketed] values are seasonality-adjusted (Fourier K=4 on day-of-year + six DOW dummies, regime dummies; Duan-smeared inverse-link prediction). For Fase I up: ISP15-win raw 5.3 vs adjusted 4.9 (winter mean is partly seasonal); DA15/ID15 raw 13.0 vs adjusted 13.9 (the reform-window jump survives deseasonalization). For TR up: ISP15-win raw 5.8 vs adjusted 5.9 (no seasonal correction needed); DA15/ID15 raw 1.8 vs adjusted 1.2 (−66% below 3-sess baseline). The TR-channel collapse is preserved after seasonality removal.

Reading the cascade. (i) **Total ajuste cost rose** under the reform — €16.1M/day (ISP15-win) → €21.1M/day (DA15/ID15), +31%. (ii) **The reform redistributed cost from TR to Fase I**: +5.7M/day in Fase I-up, −4.0M/day in TR-up — it pre-programmed predictable post-clearing Fase I redispatch and cut the real-time fixes. (iii) **TR price halved** (€545 → €226/MWh) as MTU15-IDA absorbed intra-hour scarcity; the Fase I price barely moved (€187

→ €168) but Fase I *cost* more than doubled. (iv) Seasonality control confirms the pattern and *sharpens* the TR collapse (DA15/ID15 SA TR-up €0.8M/day, a 7× reduction from ISP15-win).

Implication. MTU15 is *not* purely cost-saving — it shifts cost from real-time to pre-IDA Fase I, net + ∼ 30% per-day ancillary spending. Apparent "CCGT redispatch" under reforzada is fundamentally a pay-as-bid *Fase I* channel where southern-corridor incumbents (§7) face minimal competition; the ∼ 21M/day ajuste bill dwarfs any margin detectable from the DA auction alone, so DA-only welfare counterfactuals understate scarcity rents.

8.4 Efficiency gains across the three reform stages

The three sequential reforms (ISP15 settlement, MTU15-IDA clearing, MTU15-DA clearing) each delivered a distinct efficiency channel. The next table documents them side by side, with the outcome variable explicitly specified for each (per the never-abstract-effect rule):

Table 27: Efficiency gains per reform stage, by outcome. Reform dates: ISP15 = 2024-12-01 (settlement granularity 60-min → 15-min); MTU15-IDA = 2025-03-19 (IDA clearing 60-min → 15-min); MTU15-DA = 2025-10-01 (DA clearing 60-min → 15-min, overlaid with operational-reforzada). Per-day costs from Table 26; imbalance metrics from ESIOS indicators 762, 1338, 686, 687.

Reform	Outcome variable	Measured change
ISP15 (Dec 2024)	Mean abs imbalance vol / cell	∼810 MW (hourly, 3-sess) →∼325 MW (15-min, ISP15-win)
	Net of variance-averaging	∼50% behavioural reduction
	Dual-price spread (cobro-up / pago-dn)	40/85 → 70/110 EUR/MWh (wider, more punitive)
	DA spot price	no causal change (settlement reform, not clearing)
MTU15-IDA (Mar 2025)	TR (tiempo real) up price	€545/MWh → €226 (−59%)
	TR-up cost per day	€5.8M → €2.9M (−50%)
	DA spot price level	no clean causal change (overlap w/ reforzada)
	IDA clearing within hour	4 separate prices per hour (mechanical)
MTU15-DA (Oct 2025)	TR-up cost per day (further drop)	€2.9M → €1.8M (−38%)
	aFRR-capacity (banda) price	€13/MWh → €6/MWh (−54%)
	Fase I-up cost per day	€10.5M → €13.0M (+24% **reforzada-driven**)
	Total ajuste cost per day	€17.7M → €21.1M (+19% net)

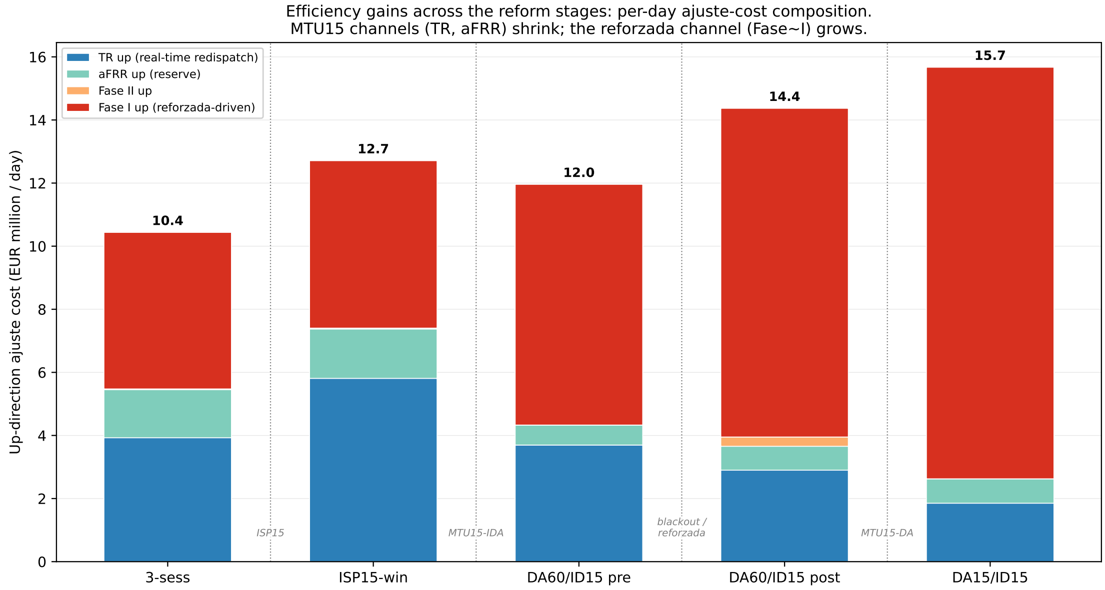


Figure 11: Per-day up-direction ajuste-cost composition across the five reform-window regimes. Stacked bars: per-day cost (EUR million) of the four up-direction channels — TR (real-time redispatch), aFRR reserve, Fase II, Fase I. Bar height = total up-direction ajuste cost per day; the number above each bar is that total. The MTU15 efficiency channels (TR up, aFRR up) shrink across the reform sequence — TR up from ~ 5.8M/day at ISP15-win to ~ 1.9M/day at DA15/ID15 — while the reforzada channel (Fase I up) grows from ~ 5M/day to ~ 13M/day. The net per-day cost rises because the reforzada-driven Fase I growth outweighs the granularity-driven TR/aFRR savings; the two are logically distinct policy effects (§6) that happen to overlap in calendar time. Window-length-normalised from `per_regime_costs_official.csv`.

Reading the efficiency gains.

- **ISP15 is the cleanest single-shock identification** (no overlapping reform in its window). The 60→15-min settlement granularity mechanically shrinks per-cell imbalance variance $\sim 4\times$, so ~ 50 pp of the observed $\sim 60\%$ reduction in mean absolute imbalance is measurement and ~ 10 pp is behavioural; the dual-price spread also widens (more punitive).
- **MTU15-IDA delivered the big TR-cost channel**: TR cost €5.8M → €2.9M/day, the largest single efficiency gain — granular IDA absorbs intra-hour scarcity that previously bled into real-time intervention.
- **MTU15-DA delivered a smaller efficiency dividend, swamped by parallel reforzada cost growth.** Clean MTU15-DA channels (further TR + aFRR savings ≈ -1.5M/day) are logically distinct from the reforzada-driven Fase I growth (+2.5M/day, tighter PO-3.x security criteria forcing more pay-as-bid CCGTs on); the two are confounded in the post-Oct-2025 window but never summed as "the effect of MTU15-DA". Separating them needs the 40-day pre-blackout window or a geographic discontinuity.

9 Cournot reading and CNMC enforcement context

9.1 Cournot reading of the cap-padded ladder

A near-constant DA bid price ladder at the cap (GN at ~ 840 EUR/MWh in Table 23) combined with all the adjustment on the *quantity* side — the unit either clears DA or it does not, but its bid price hardly varies — is the empirical signature of a **Cournot strategy on the quantity margin**. In pure economic terms this is equivalent to direct price-raising in DA, but the regulatorily defensible posture for the firm is "my offer price has not changed; my bids follow my published ladder." The redispatch channel (Fase I pay-as-bid) is then where the per-MWh rent is captured.

9.2 Historical CNMC sanction precedent

The empirical pattern documented in §§5–7 has a direct precedent in CNMC enforcement. The resolutions for the relevant cases are archived in the project's regulation document folder.

Table 28: CNMC sanction resolutions relevant to the practice. "LSE" = Ley del Sector Eléctrico 24/2013. Sources: CNMC resolution PDFs archived in the project regulation folder.

Year	Firm / Plant	Expediente	LSE article	Sanction
2015	Iberdrola hydro	SNC-DE-046-14	64 muy grave (manipulation)	25.0 M EUR
2019	**Naturgy CCGT**†	SNC-DE-175-17	65.34 grave (abnormal DA offers)	∼7 M EUR
2019	Endesa Besós 3	SNC-DE-174-17	65.34 grave (abnormal DA offers)	6.6 M EUR
2023	Naturgy Sabón 3 (Galicia)	—	65.33 grave (RT abuse)	confidential
2023	Engie Castelnou	SNC-DE-152-22	grave (availability)	—
2023	Ignis Escatrón 2	SNC-DE-151-22	grave (availability)	—

† Plants in scope: Besós 4, PBCN 1/2, Sagunto 1/2/3, Málaga 1, SROQ 1 — exactly the Cataluña / Levante / Sur zonal map (§7). CNMC-estimated minimum benefit 13.0 M EUR over 4 months.

Reading. The 2019 Naturgy sanction (SNC-DE-175-17) is the smoking-gun precedent: same plants, same zones, same mechanism (high DA → unit not despatched in DA → unit systematically programmed at higher RT prices) over October 2016 – January 2017. The 2024–26 evidence in this Part B is the same practice continuing post-*operación reforzada*, now at fleet-wide scale, with the within-hour granularity reform as an additional tool. The 2023 Sabón 3 sanction extends the geography to Galicia (the GE / GN duopoly per §7).

9.3 Blackout-cohort expedientes (separate channel)

In April 2026 CNMC opened a parallel set of ∼ 56 expedientes (IB 22, GE 17, GN 5, others) under Art 65.8 of the LSE, for **voltage-control violations** on April 28, 2025 (the day of the blackout). This is a *distinct regulatory channel*: it concerns failure to provide reactive support during the blackout sequence, not auction-shaping in DA. We flag it for completeness; the two channels are not conflated here.